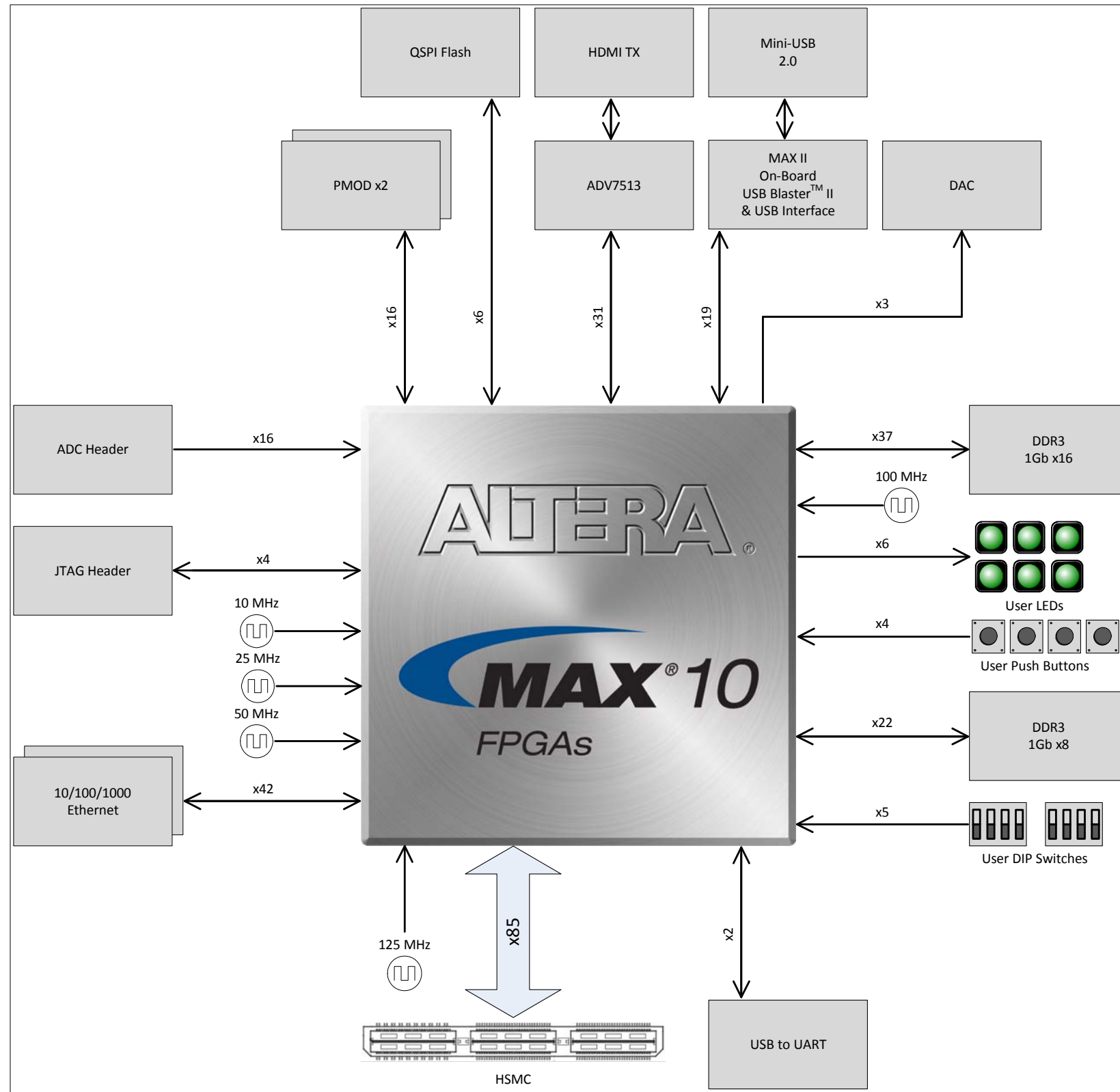


1. Project Drawing Numbers:

Raw PCB	100-0321401- C1
Gerber Files	110-0321401- C1
PCB Design Files	120-0321401- C1
Assembly Drawing	130-0321401- C1
Fab Drawing	140-0321401- C1
Schematic Drawing	150-0321401- C1
PCB Film	160-0321401- C1
Bill of Materials	170-0321401- C1
Schematic Design Files	180-0321401- C1
Functional Specification	210-0321401- C1
PCB Layout Guidelines	220-0321401- C1
Assembly Rework	320-0321401- C1

MAX 10 Development Kit Board

[illegible]

PAGE	DESCRIPTION	PAGE	DESCRIPTION
1	Title, Notes, Block Diagram, Rev. History	30	MAX10 Ground
2	Power Tree	31	Decoupling
3	Clock Tree		
4	MAX10 Bank 1 & 2		
5	MAX10 Bank 3 & 4		
6	MAX10 Bank 5 & 6		
7	MAX10 Bank 7 & 8		
8	MAX10 Configuration		
9	MAX10 Clocks		
10	PLL		
11	ADC Filter		
12	DAC		
13	DDR3 SDRAM		
14	QSPI FLASH		
15	HSMC Port		
16	GPIO, PMOD		
17	HDMI		
18	10/100/1000 Ethernet A		
19	10/100/1000 Ethernet B		
20	USB to UART		
21	On-Board USB Blaster II-1		
22	On-Board USB Blaster II-2		
23	LED, User IO, Connector		
24	Power1		
25	Power2		
26	Power3		
27	Power4		
28	Power5		
29	MAX10 Power		



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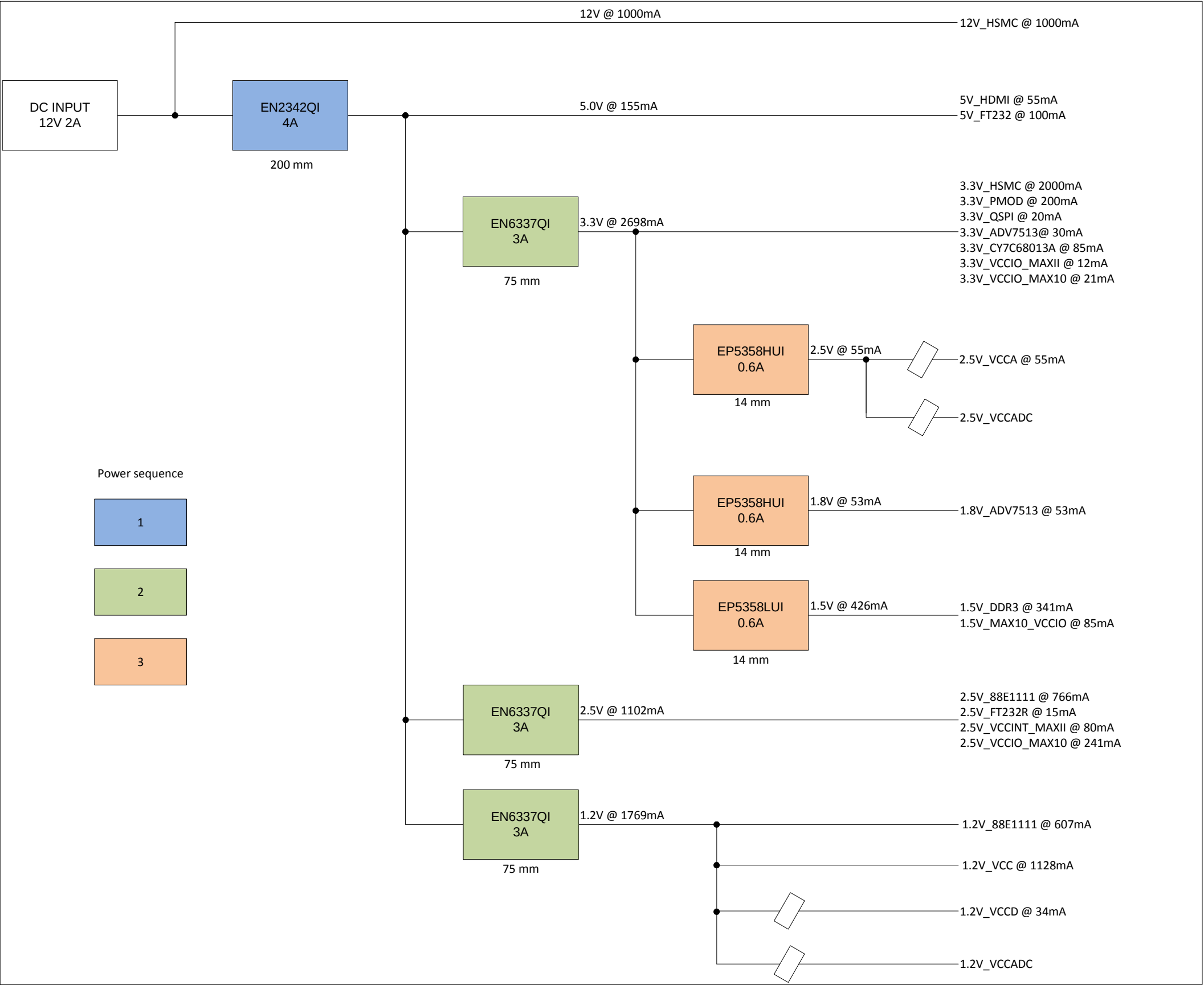
Title	MAX10 FPGA Development Kit (6XX-44292R)
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Size B	Document Number 150-0321401-C1
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Rev	
C1	

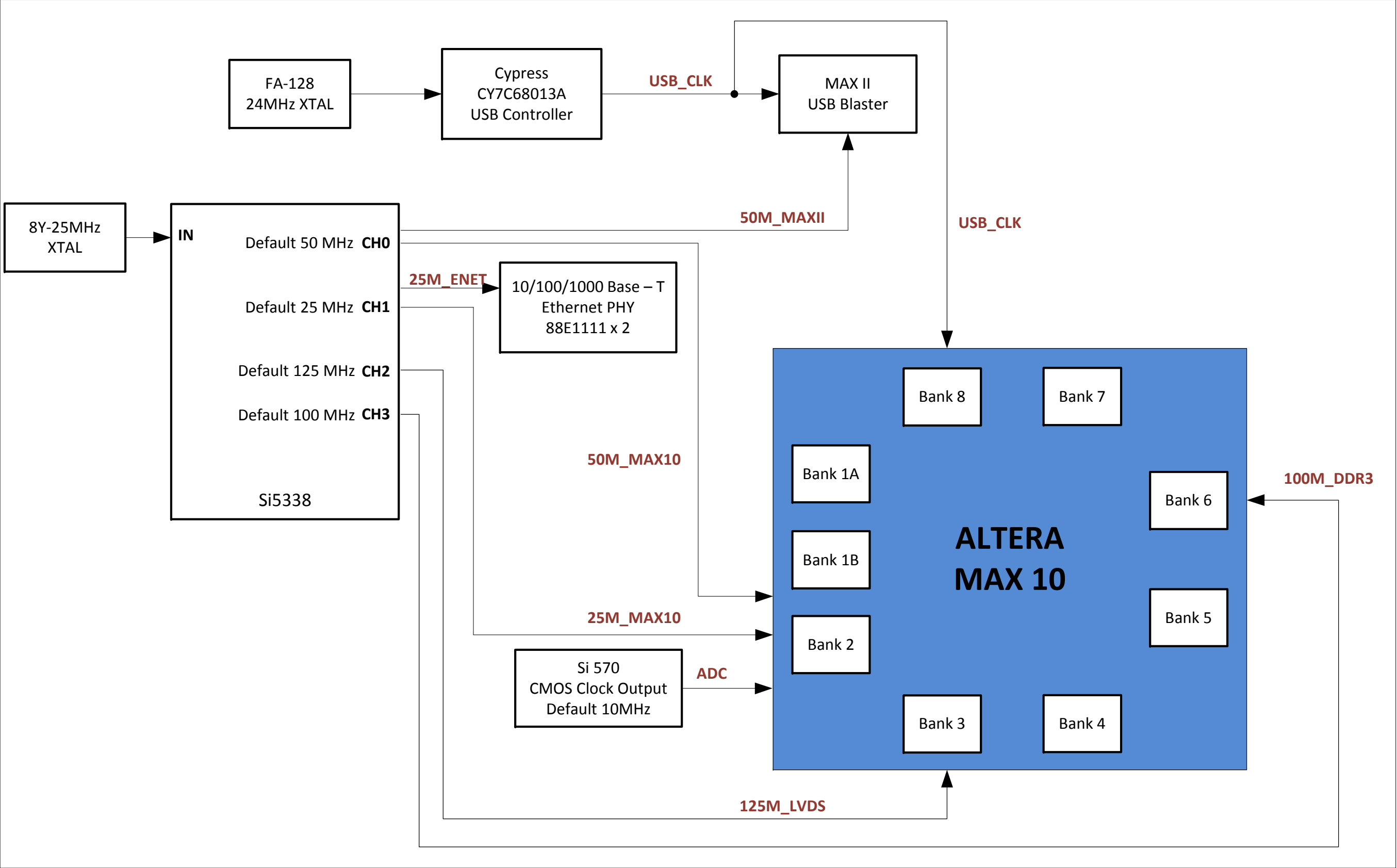
Date: Wednesday, March 18, 2015 Sheet 1 of 31

Power Tree



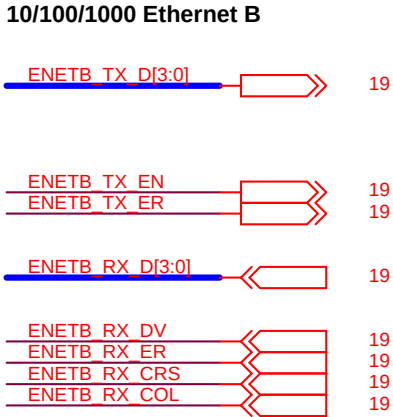
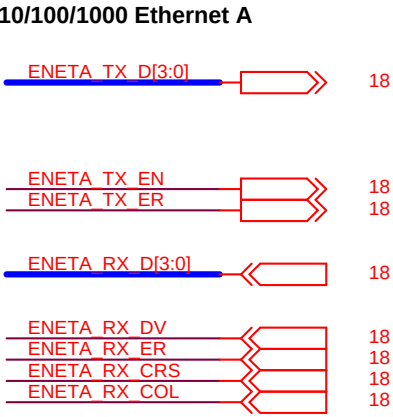
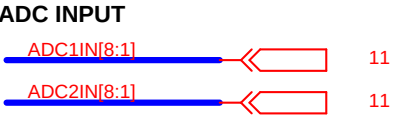
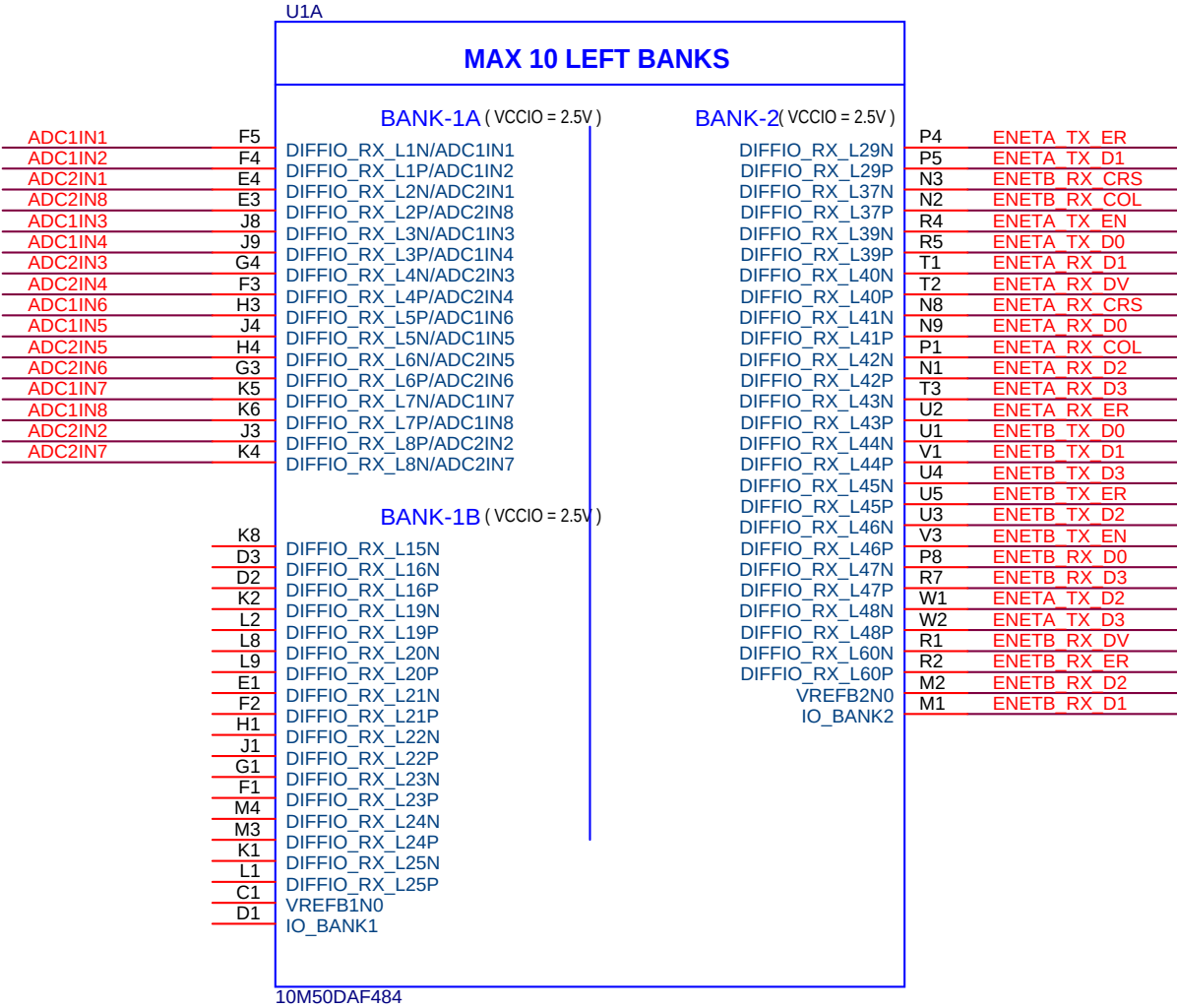
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Title MAX10 FPGA Development Kit (6XX-44292R)		
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Clock Tree

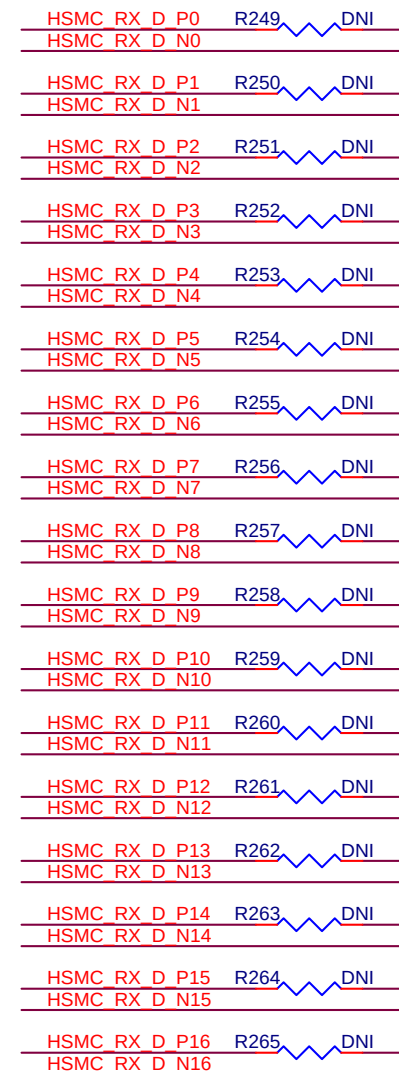
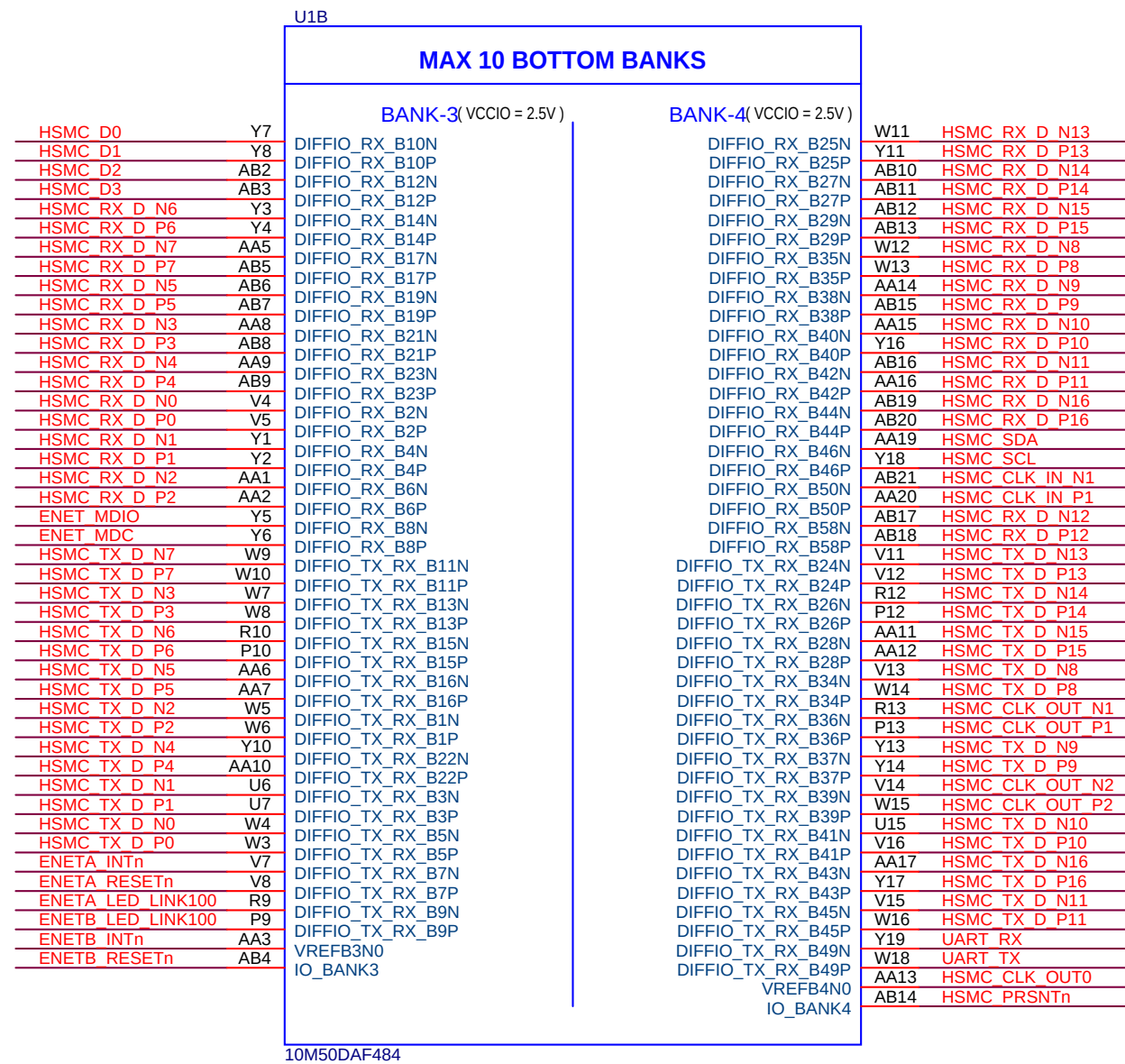


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MAX10 Bank 1 & 2

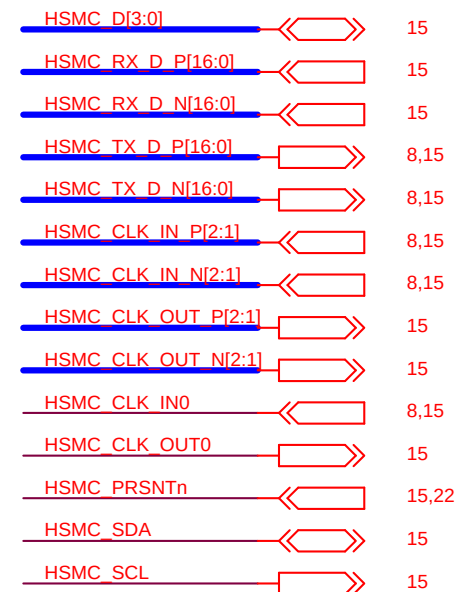


MAX10 Bank 3 & 4

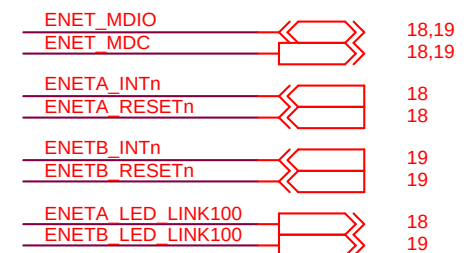


Notes:
R249-R265 are 100 ohm termination resistors for LVDS RX.

HSMC Interface



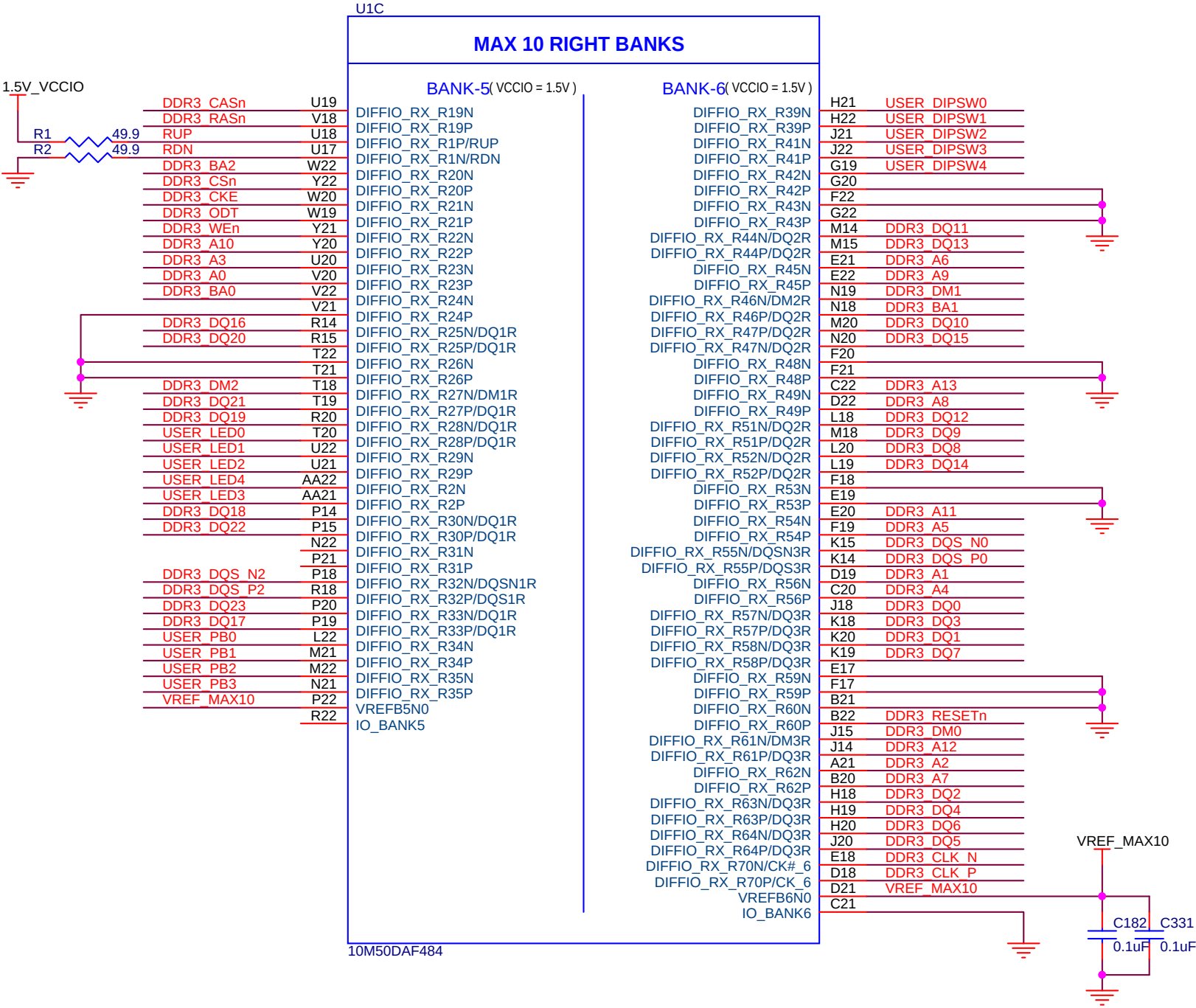
Ethernet



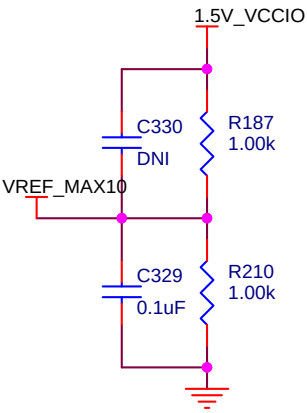
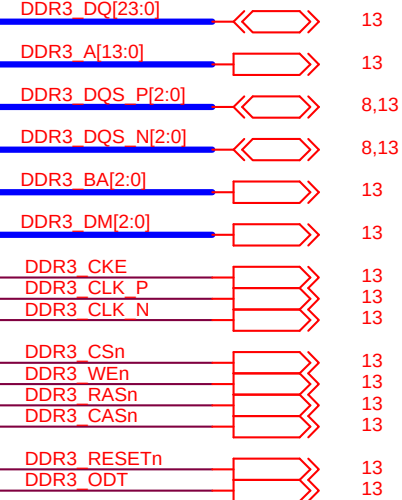
UART



MAX10 Bank 5 & 6

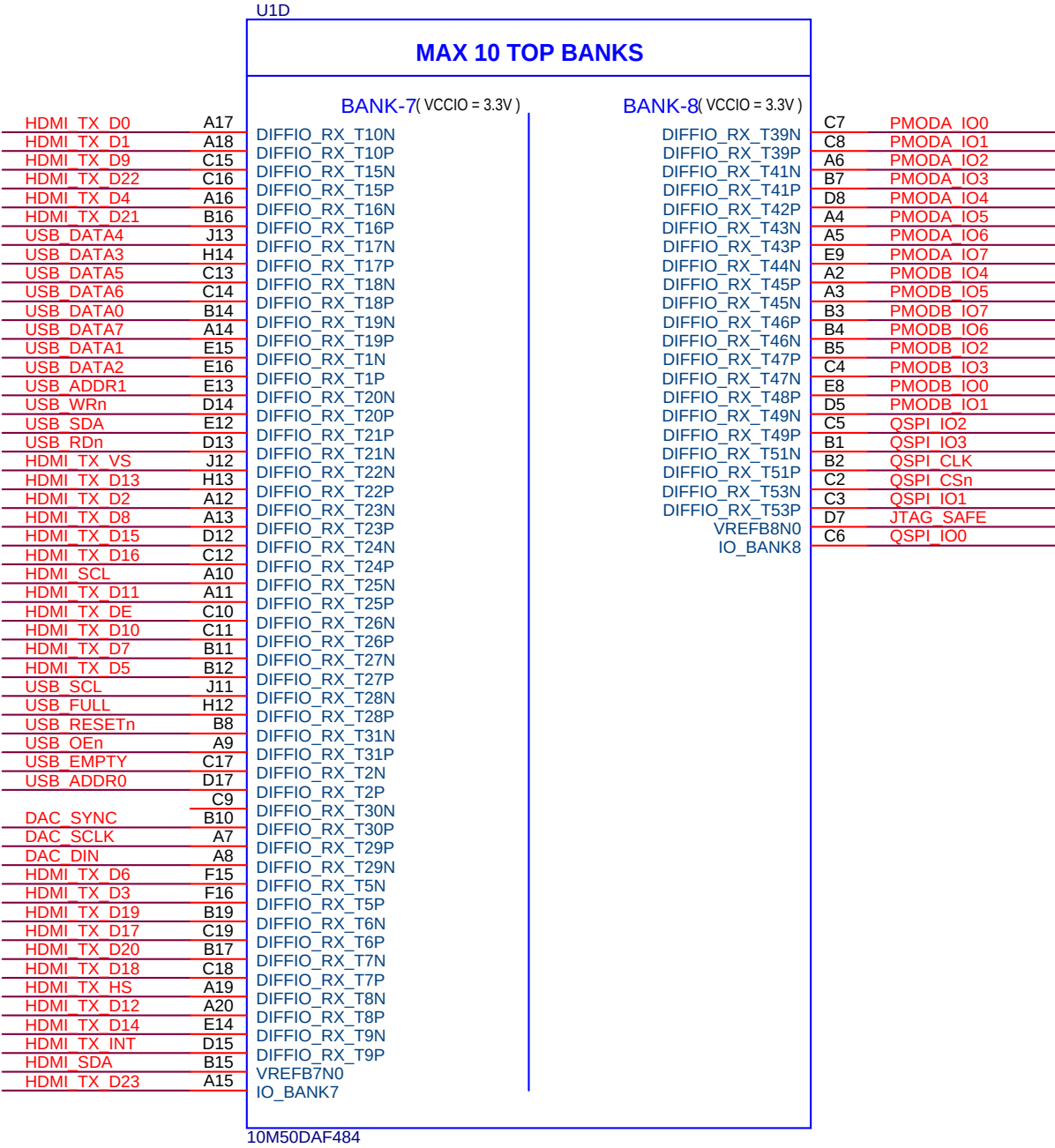


DDR3 Interface

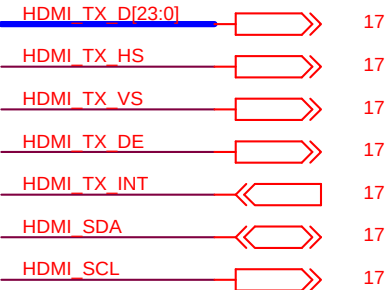


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Title MAX10 FPGA Development Kit (6XX-44292R)		
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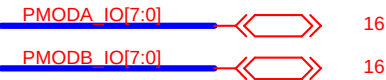
MAX10 Bank 7 & 8



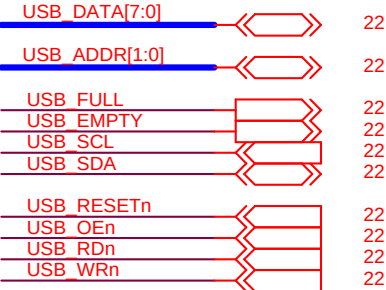
HDMI TX



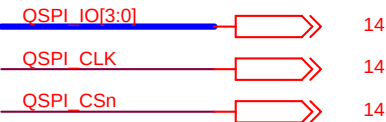
PMOD



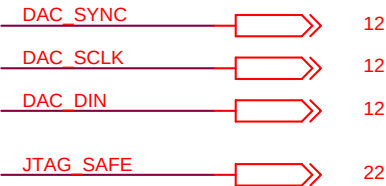
MAX10 USB INTERFACE



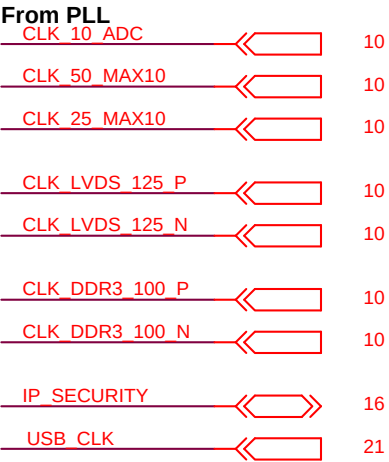
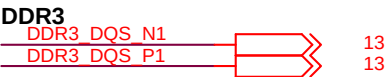
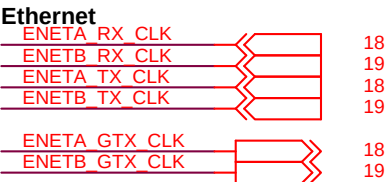
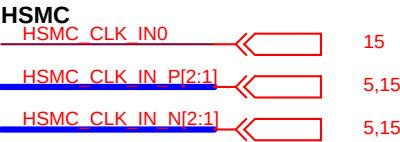
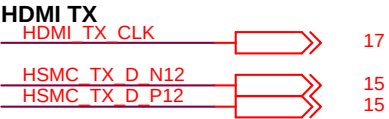
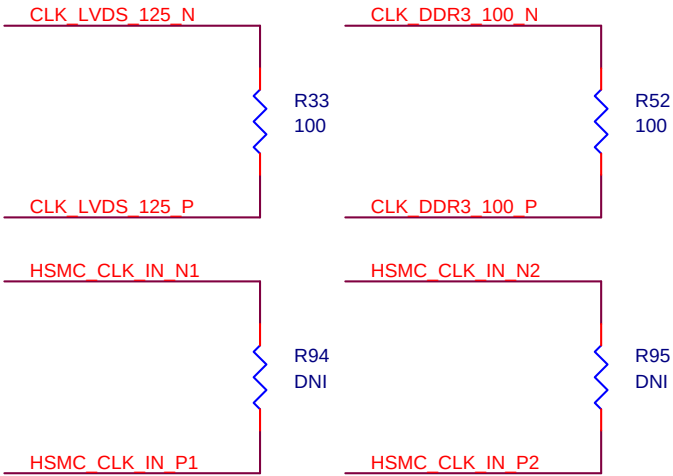
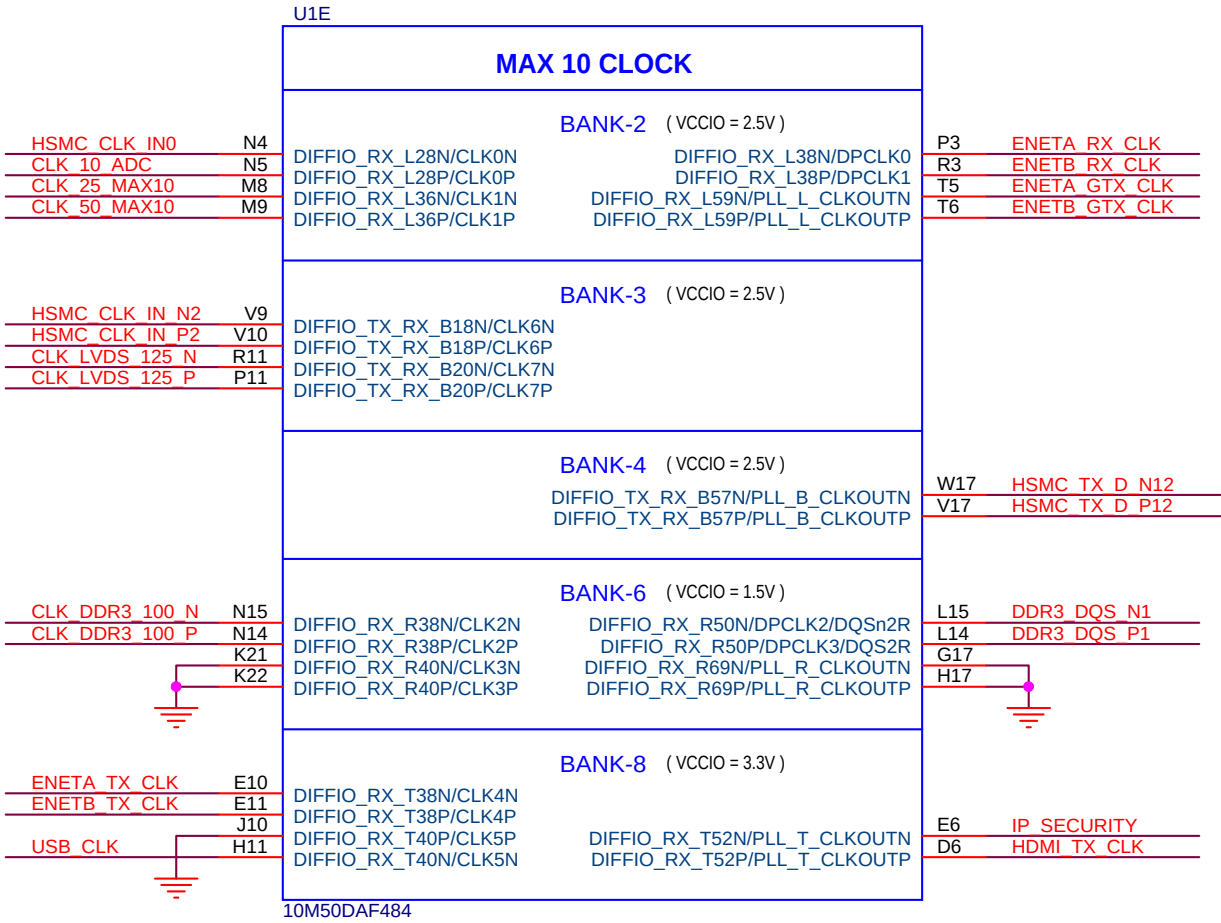
QSPI FLASH



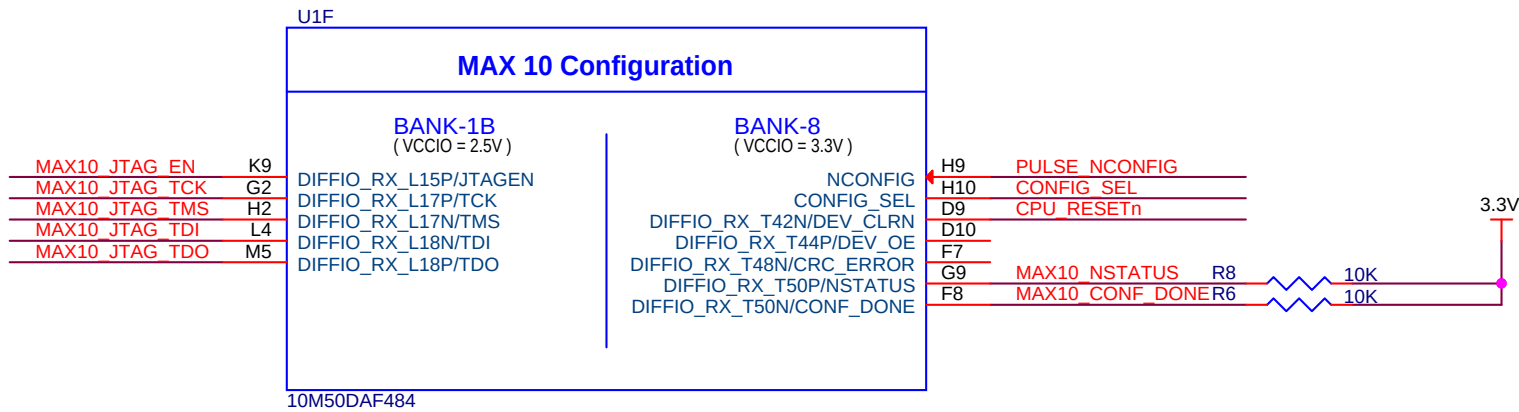
DAC



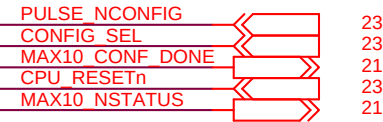
MAX10 Clock



MAX10 Configuration



Configuration



JTAG



3



Notes:
Use Clock Control GUI
Default 10MHz
I2C Address 55 HEX



Notes:
Si5338 Programmable Oscillator Use Clock Control GUI
(Defaults 50MHz, 25MHz, 125MHz, 100MHz)
I2C Address 70 HEX

Clock control

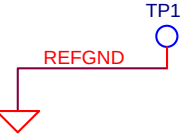
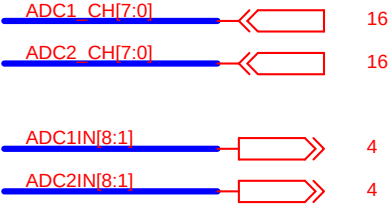
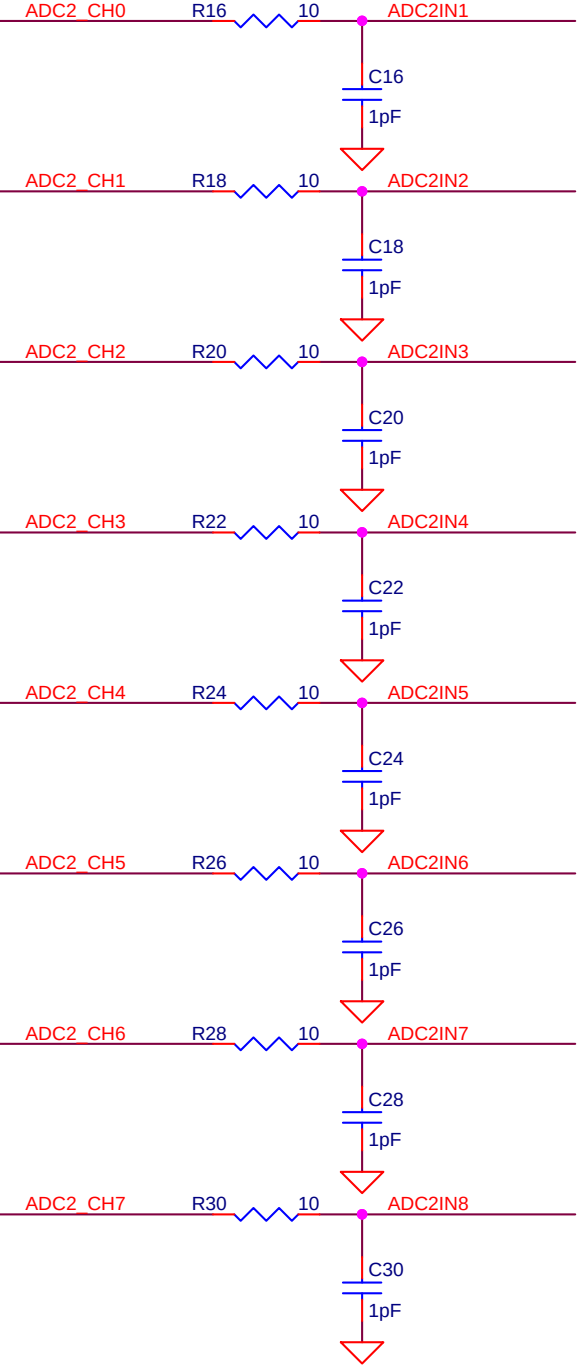
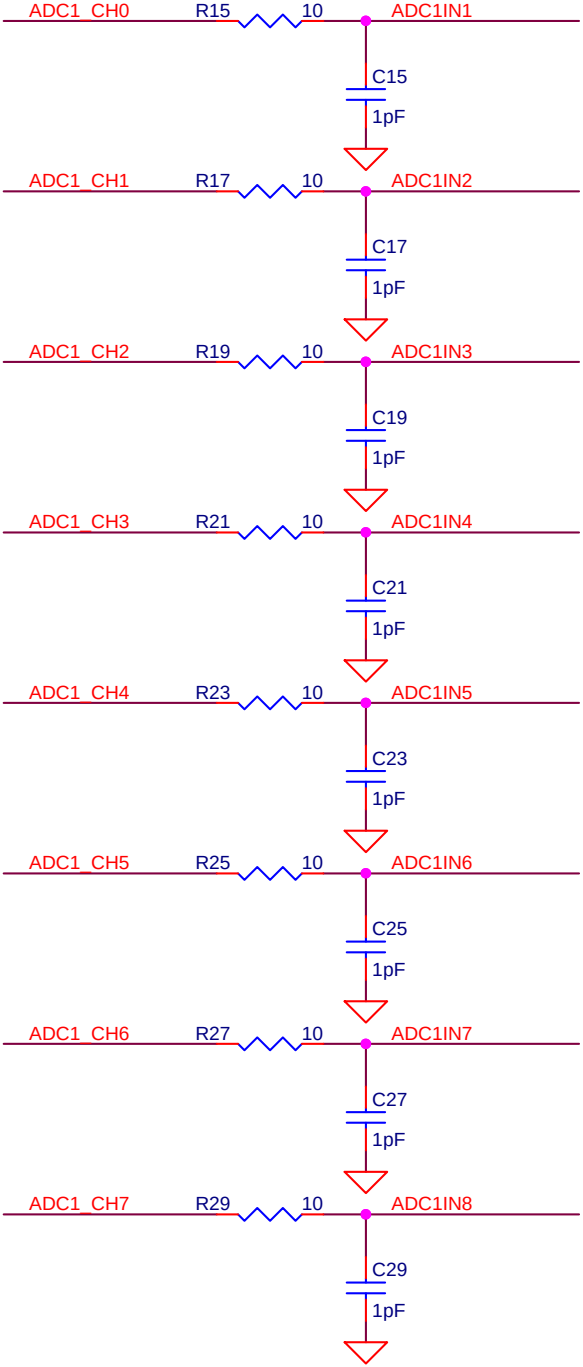
I2C_MAXII_SCL	22,28
I2C_MAXII_SDA	22,28
SI5338A_INTR	22
SI570_EN	22

Clock out

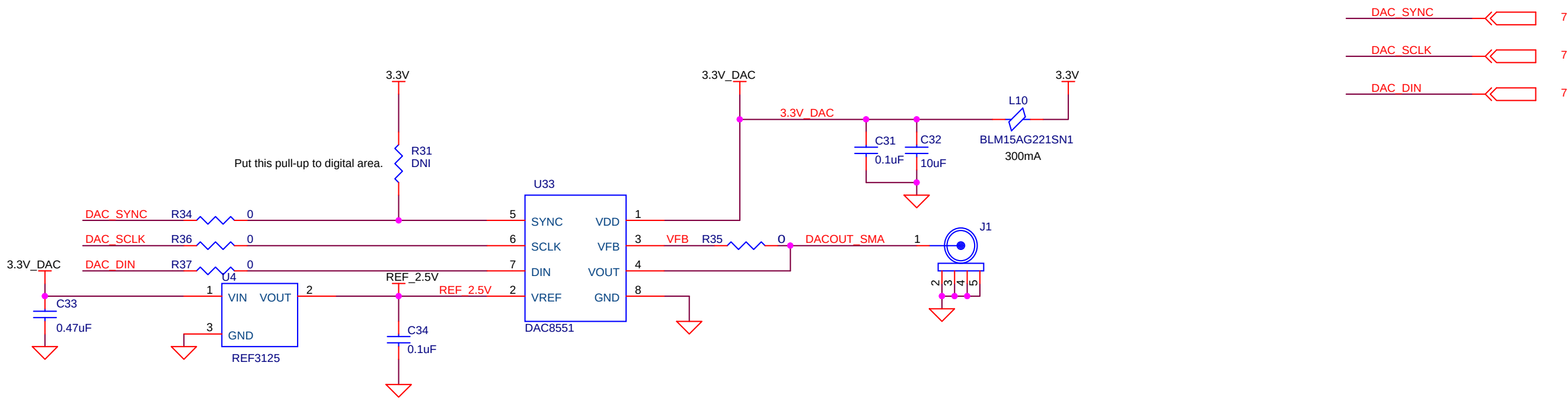
CLK 10 ADC	8
CLK 50 MAX10	8
CLK 50 MAXII	21
CLK 25 ENET	18,19
CLK 25 MAX10	8
CLK LVDS 125 P	8
CLK LVDS 125 N	8
CLK DDR3 100 P	8
CLK DDR3 100 N	8

ADC Filter

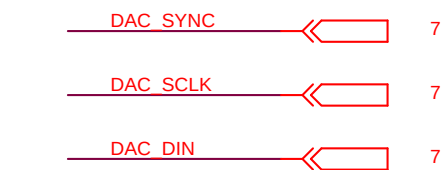
Notes:
Put the 1pF capacitors close to each MAX10 analog pin.



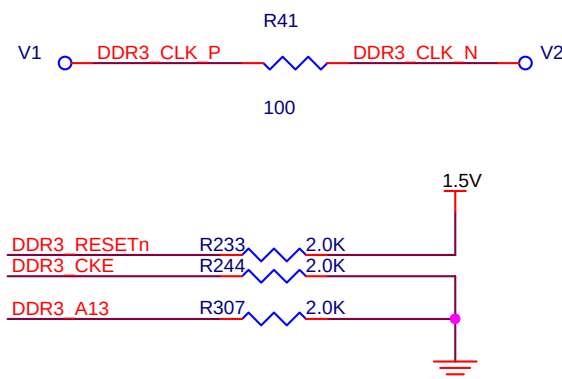
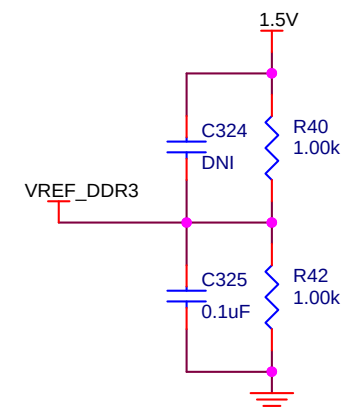
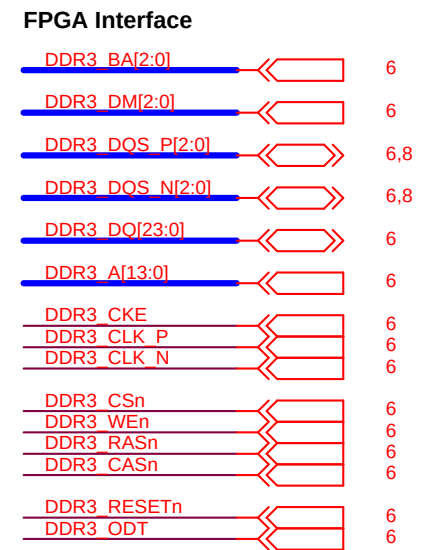
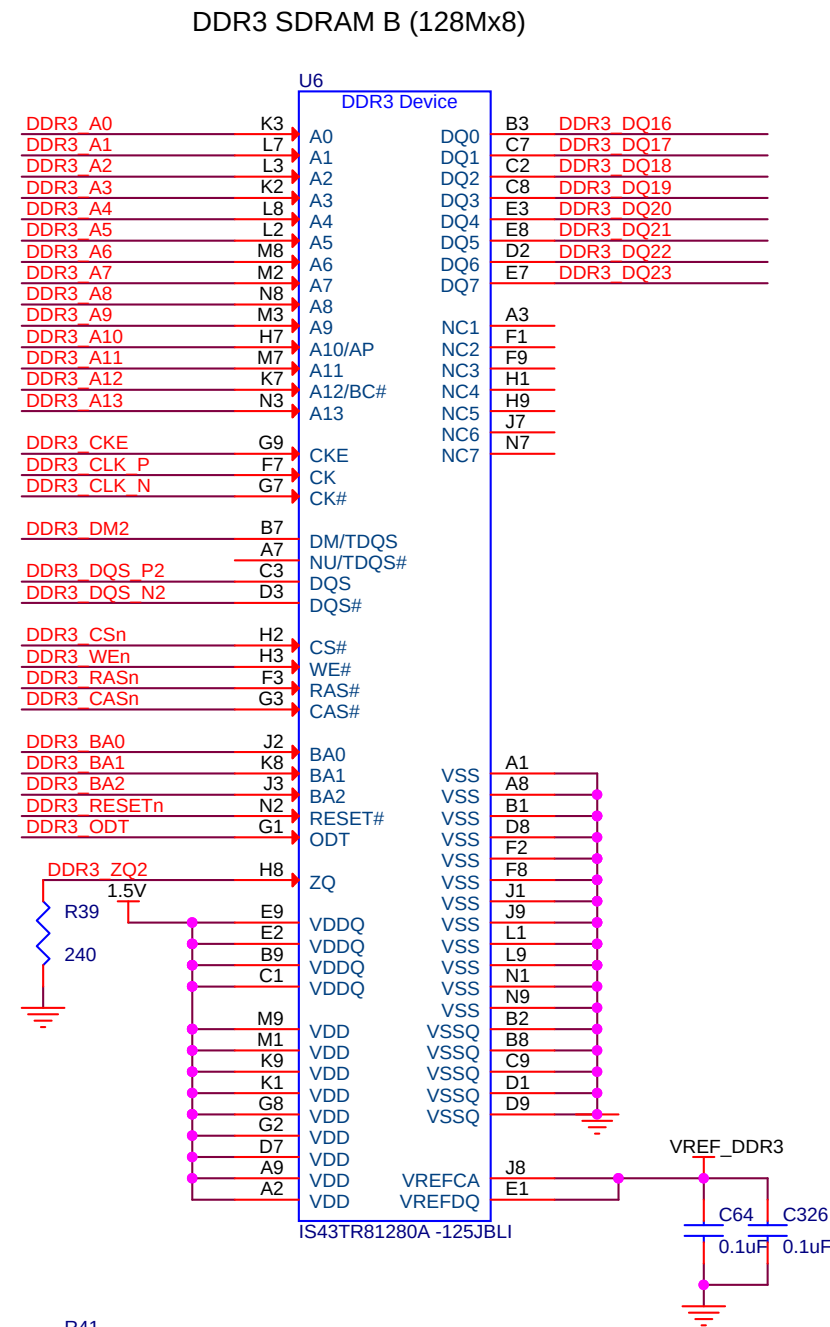
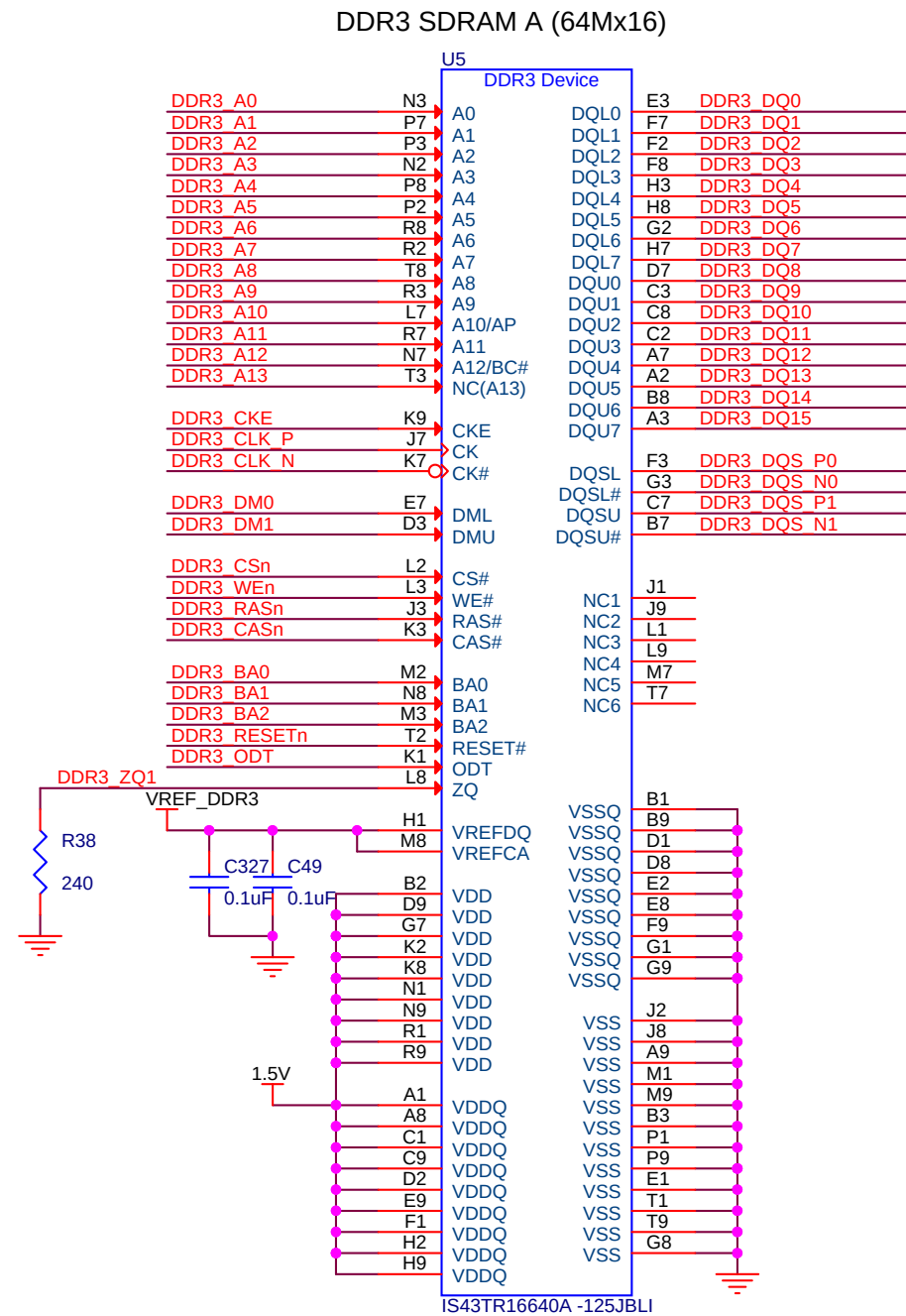
DAC



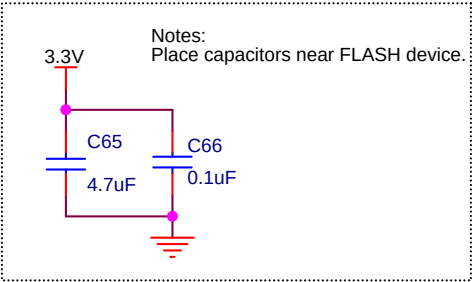
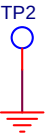
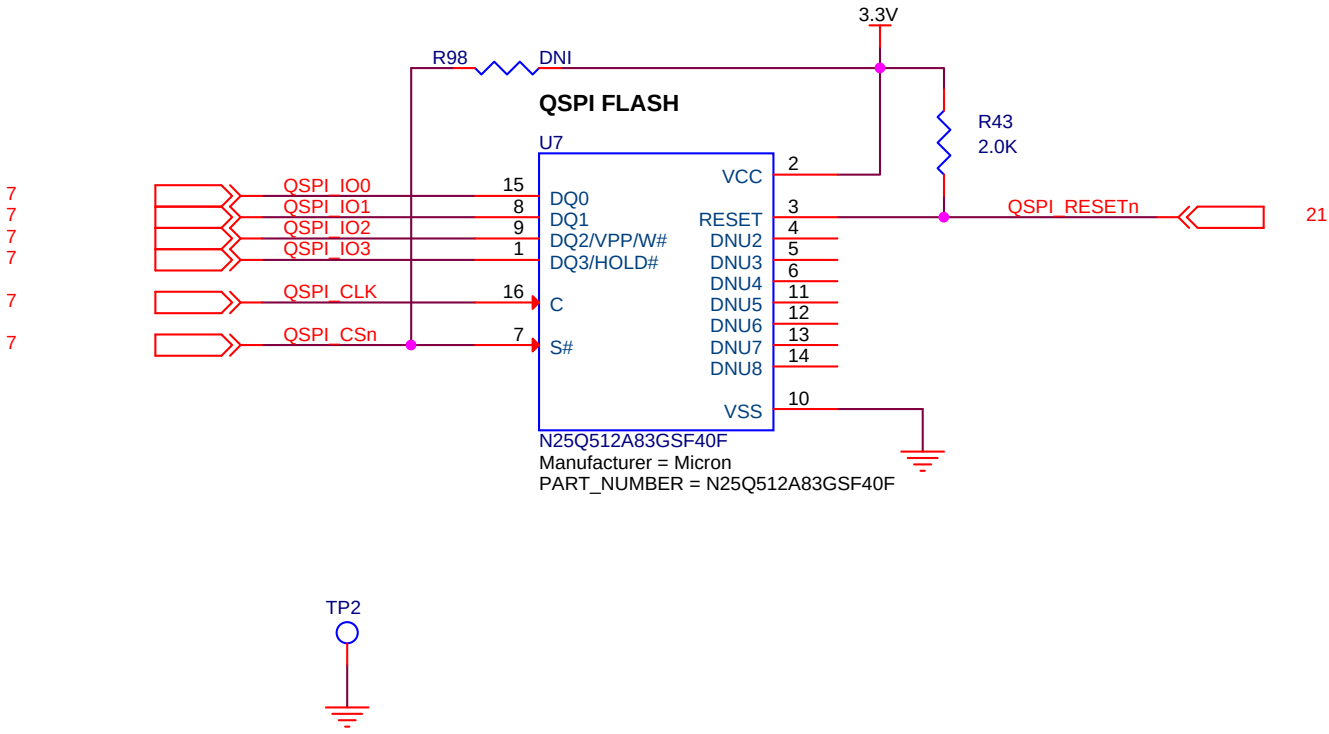
Notes:
1. Put 0.1uF capactor close to VREF pin.
2. Everything related to this DAC should be connected to analog ground.
3. For better DC accuracy, VFB should be connected to VOUT at loadpoints.
4. Thermal pad should be connected to GND.



DDR3 SDRAM



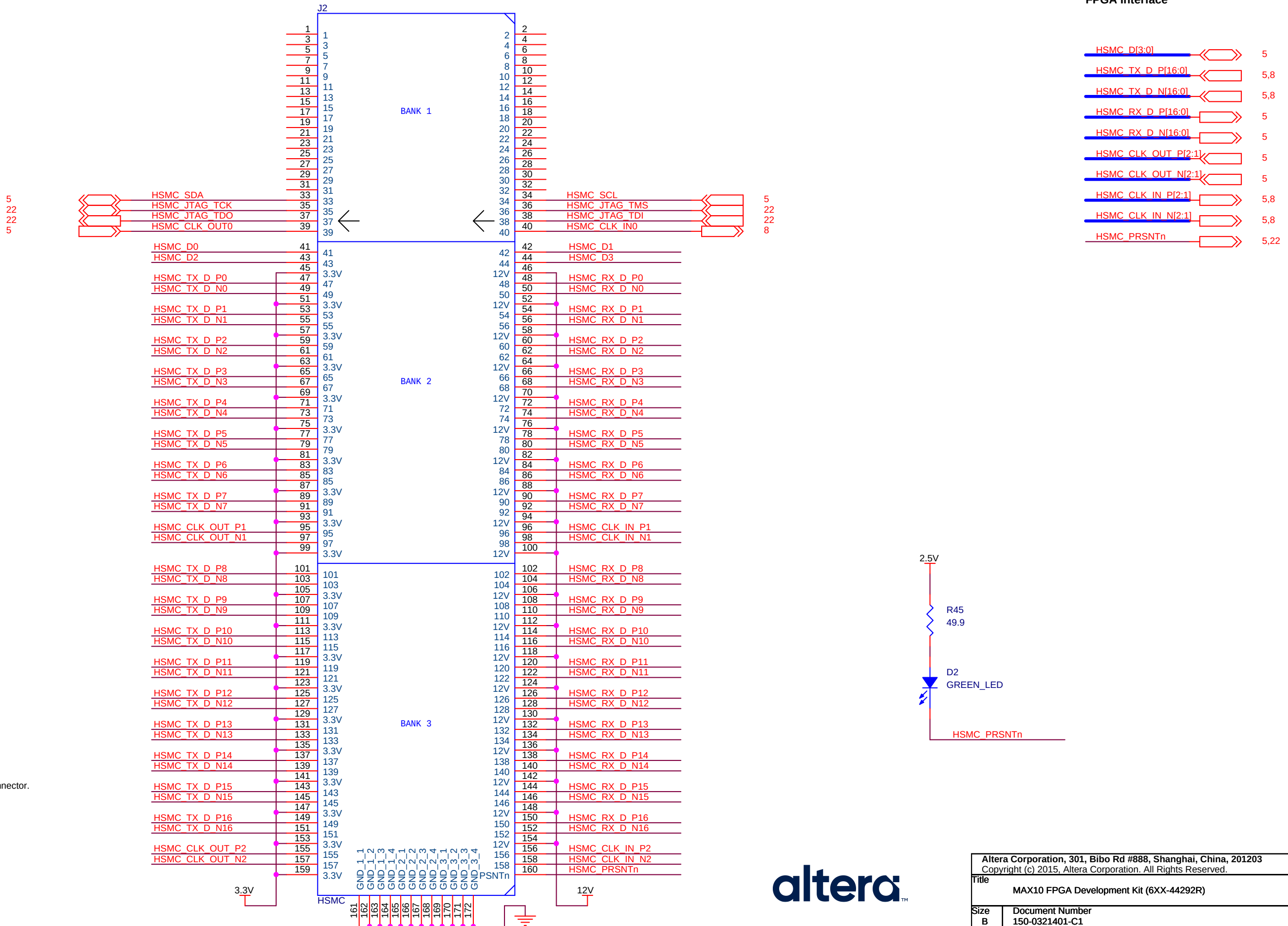
QSPI FLASH



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HSMC Port

FPGA Interface

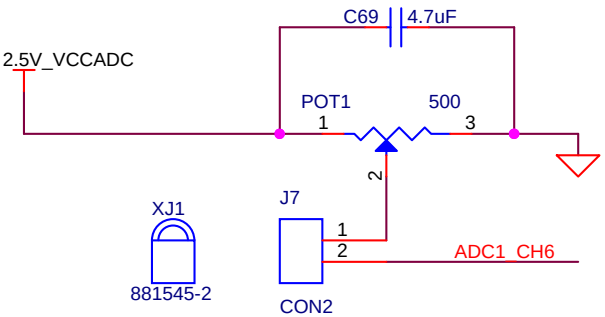
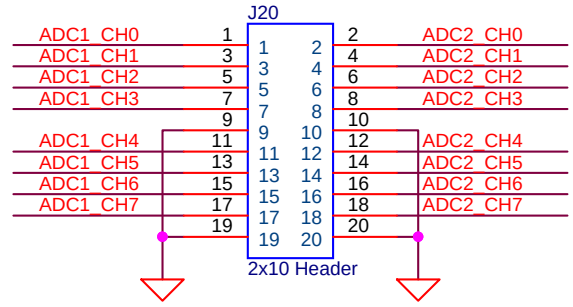
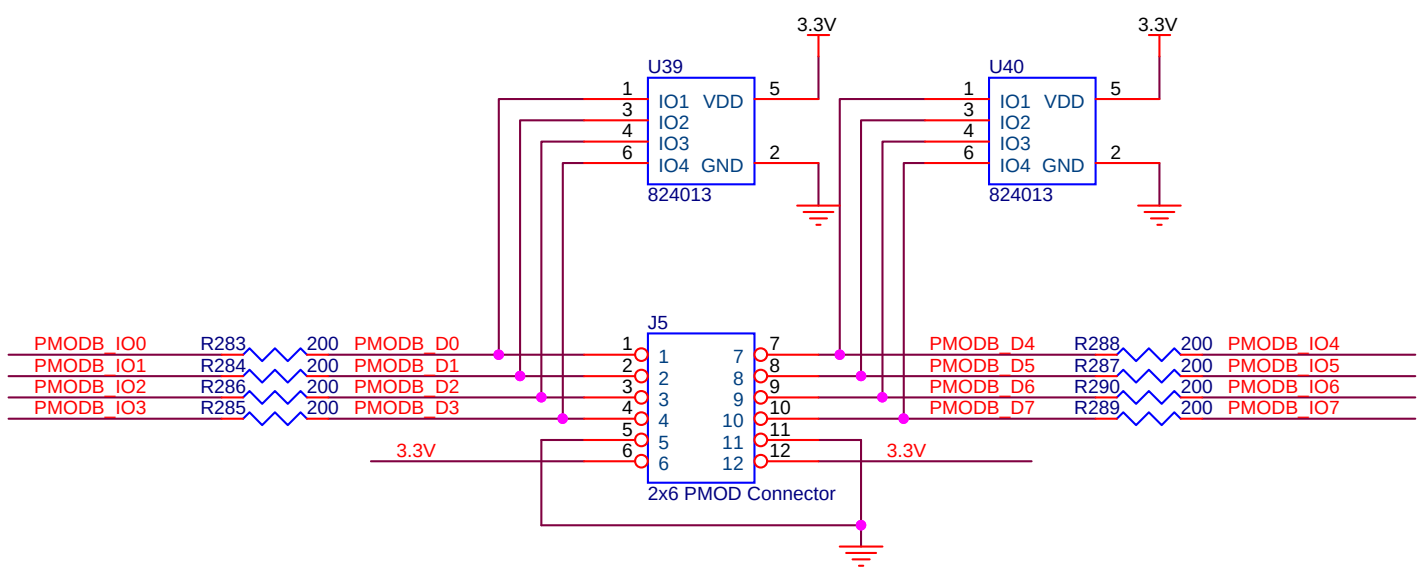
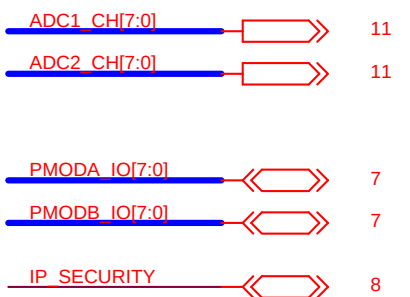
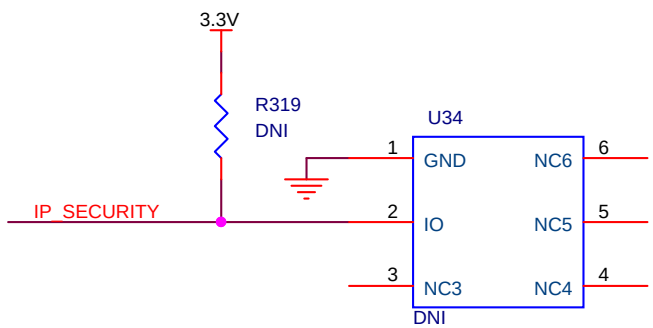
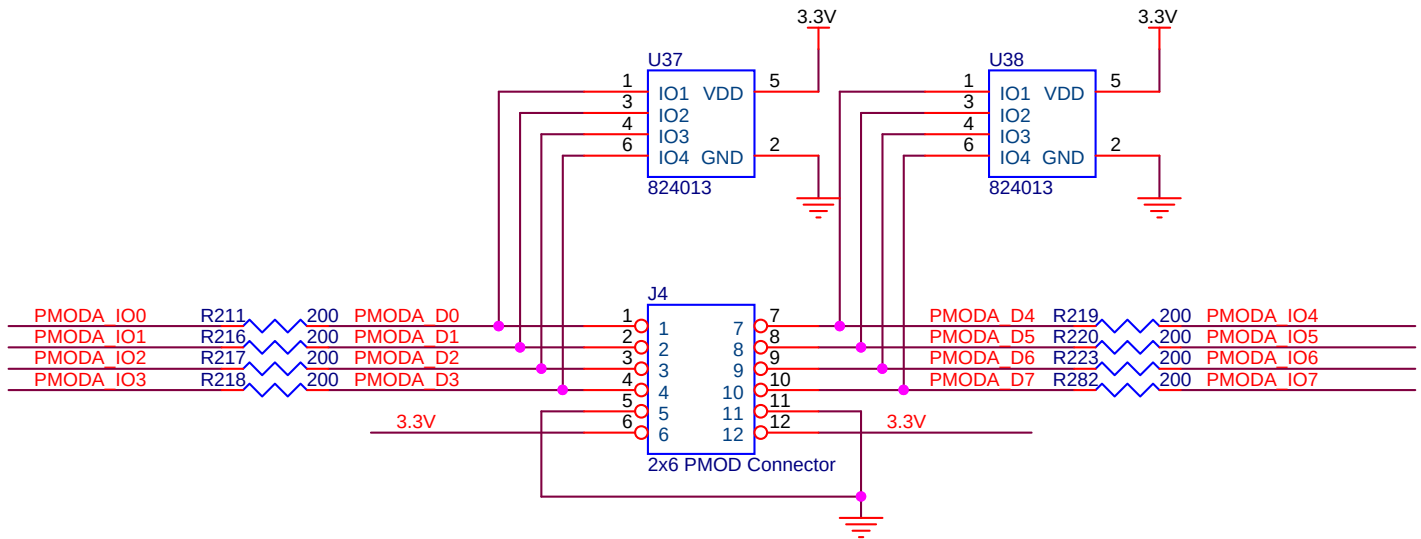


Note:
Place capacitors near HSM connector.

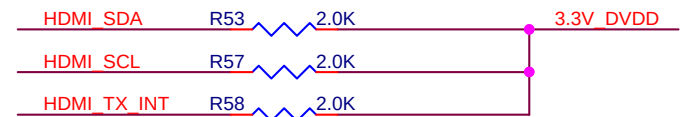
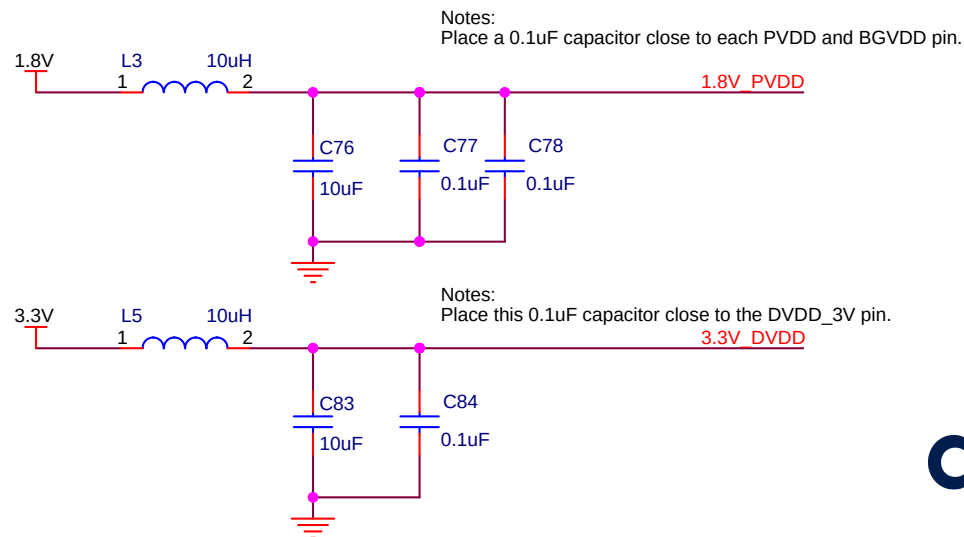
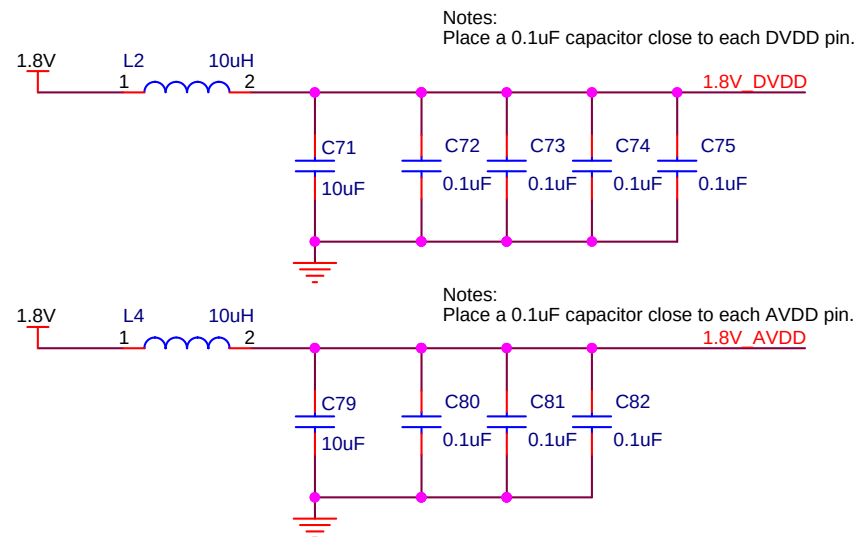
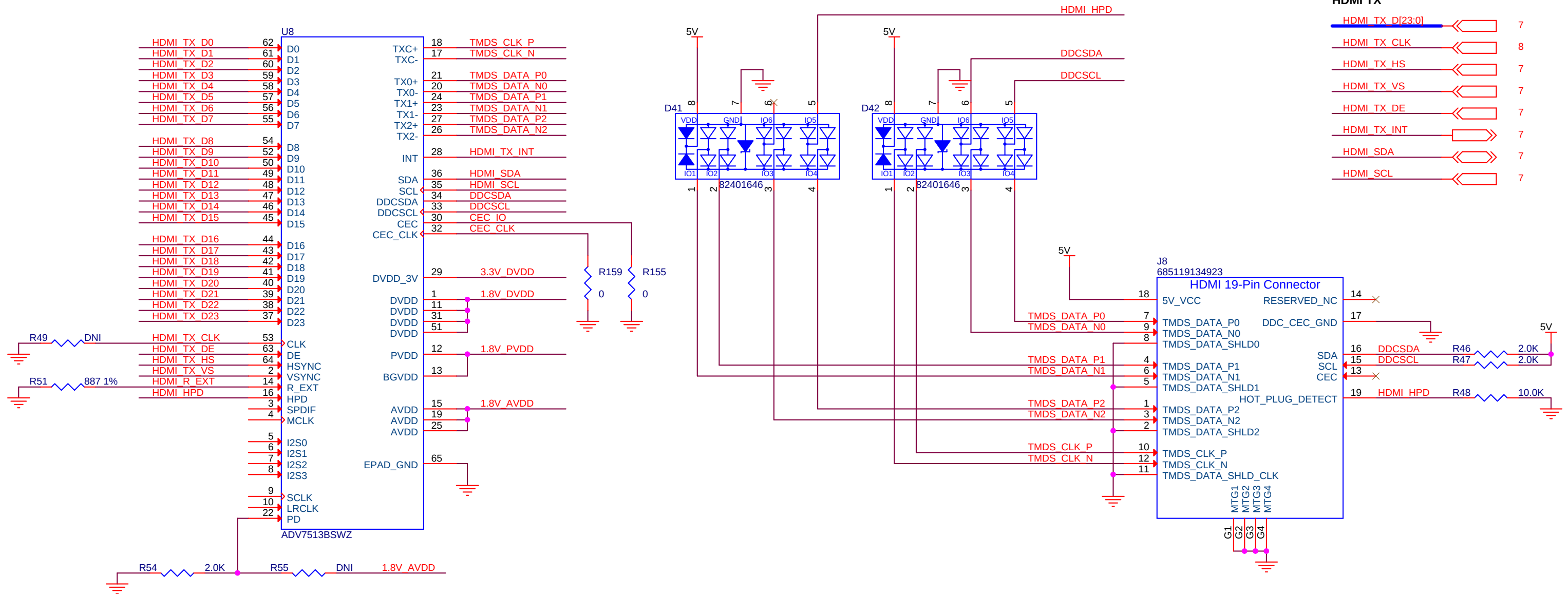


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GPIO, PMOD

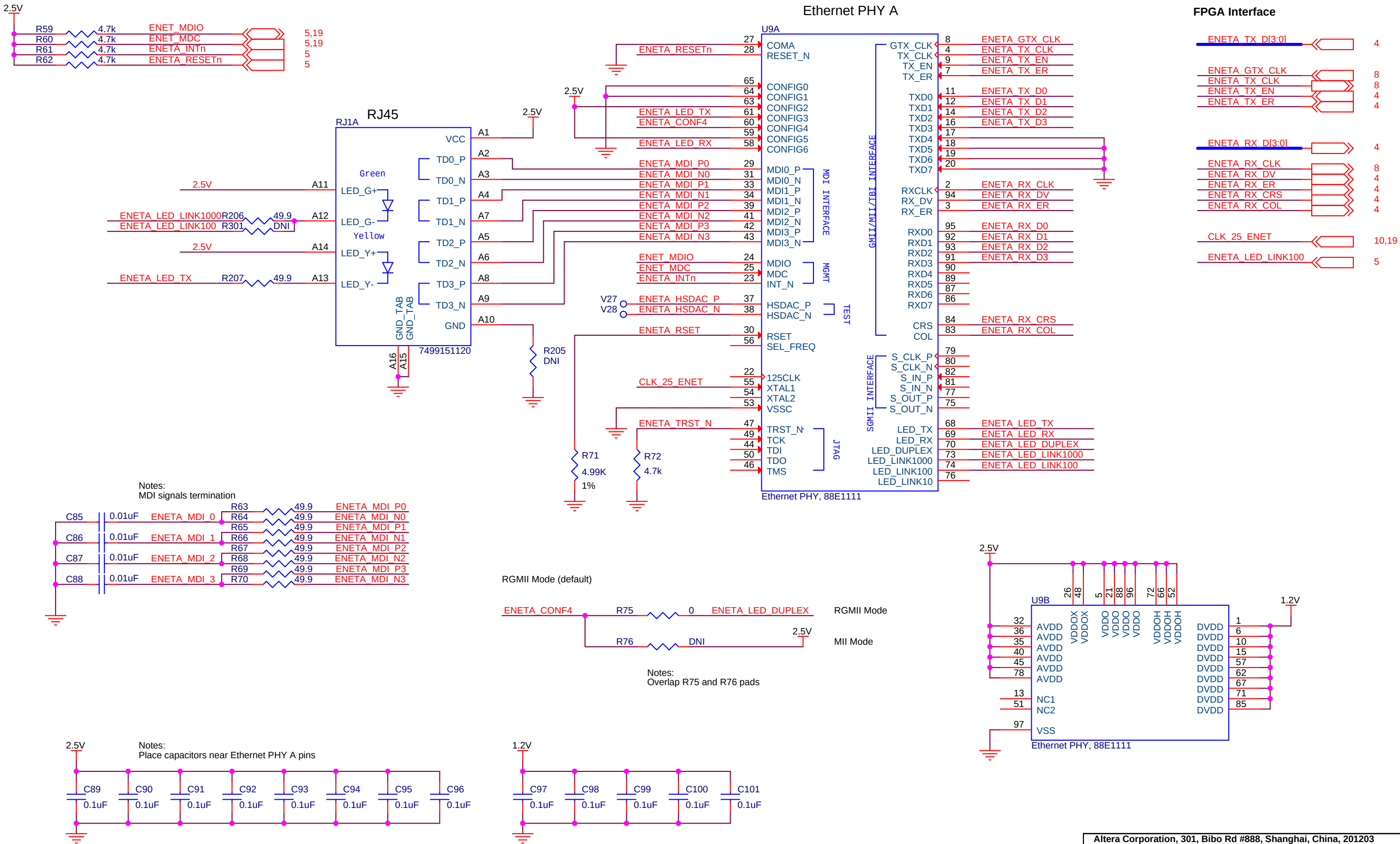


HDMI

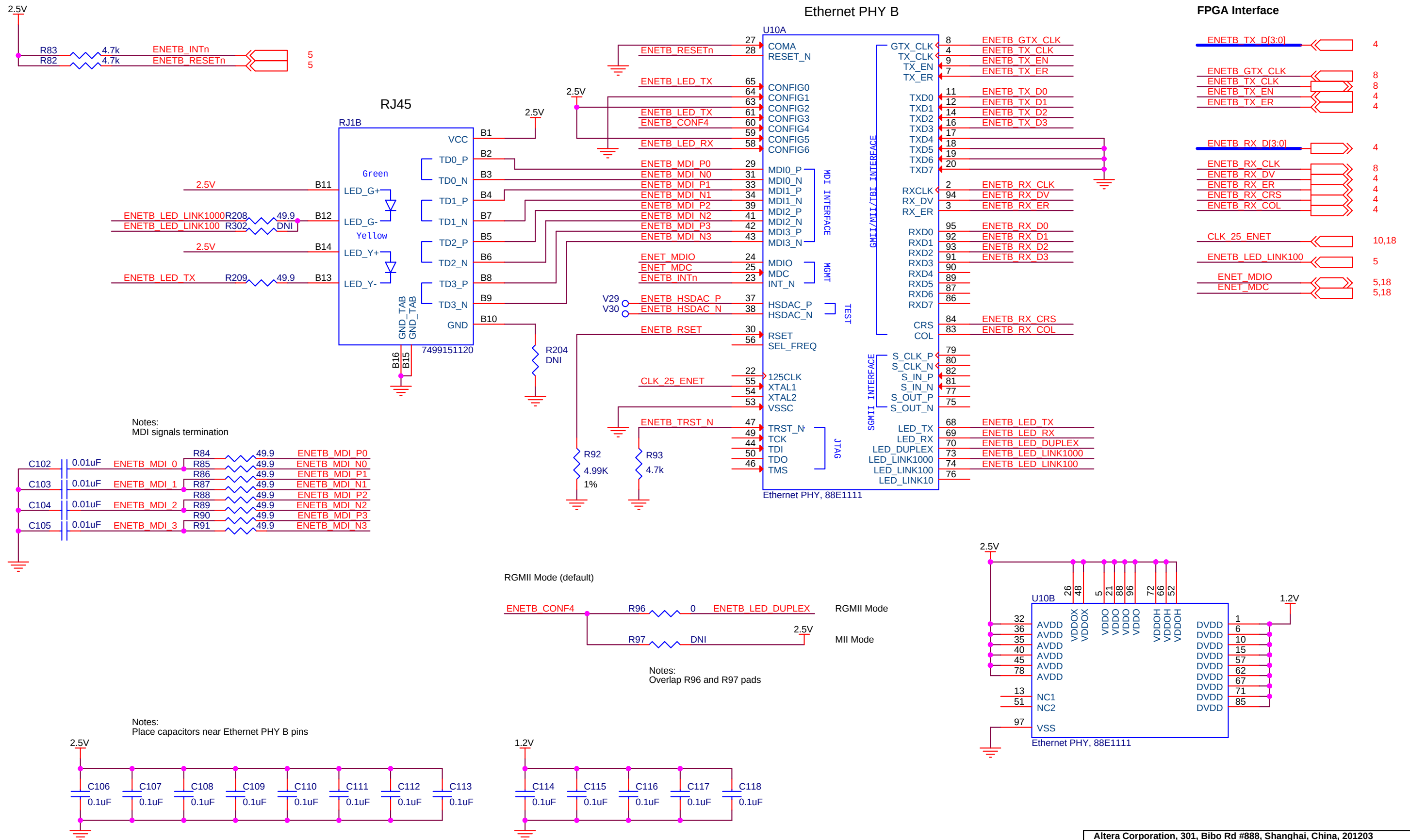


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10/100/1000 Ethernet A



10/100/1000 Ethernet B

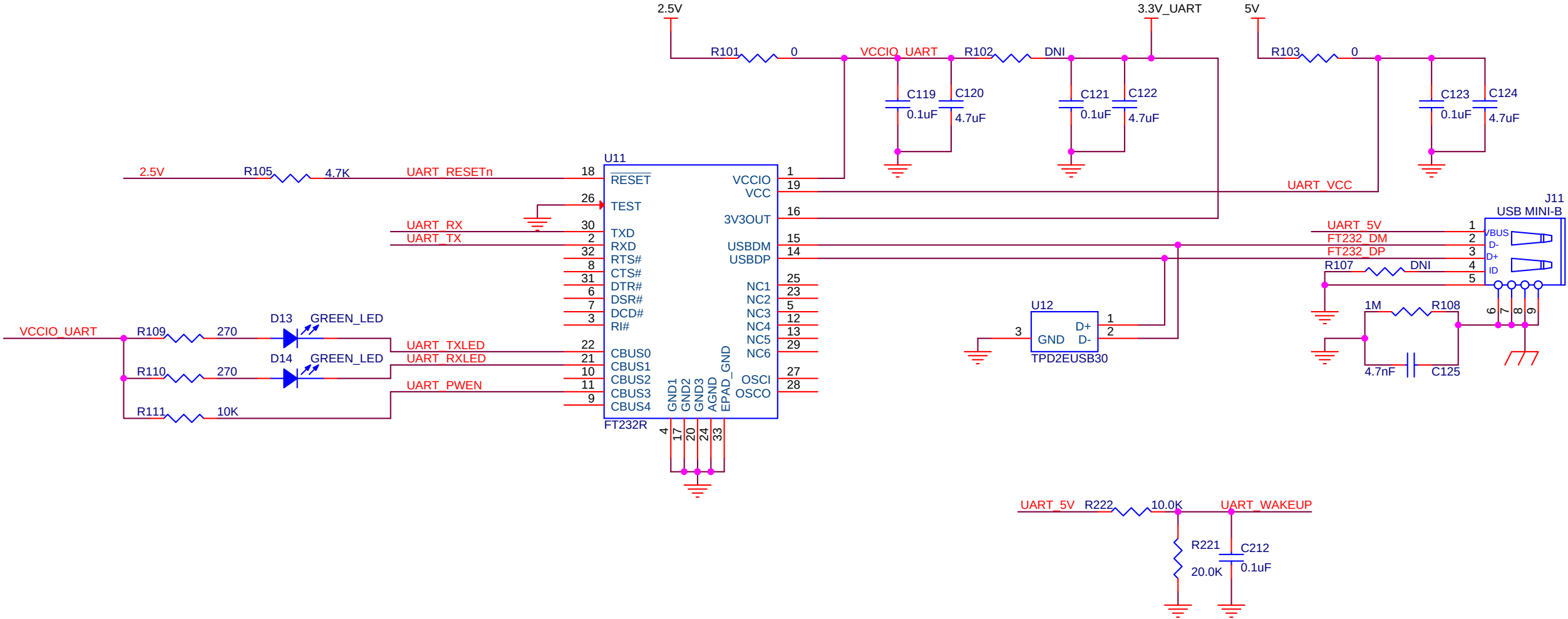


USB to UART

FPGA Interface

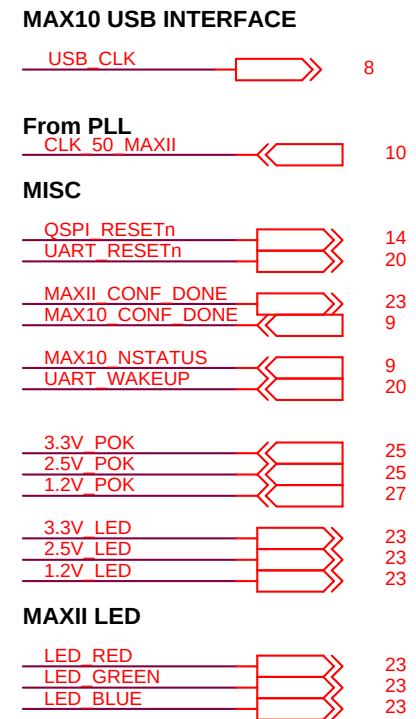
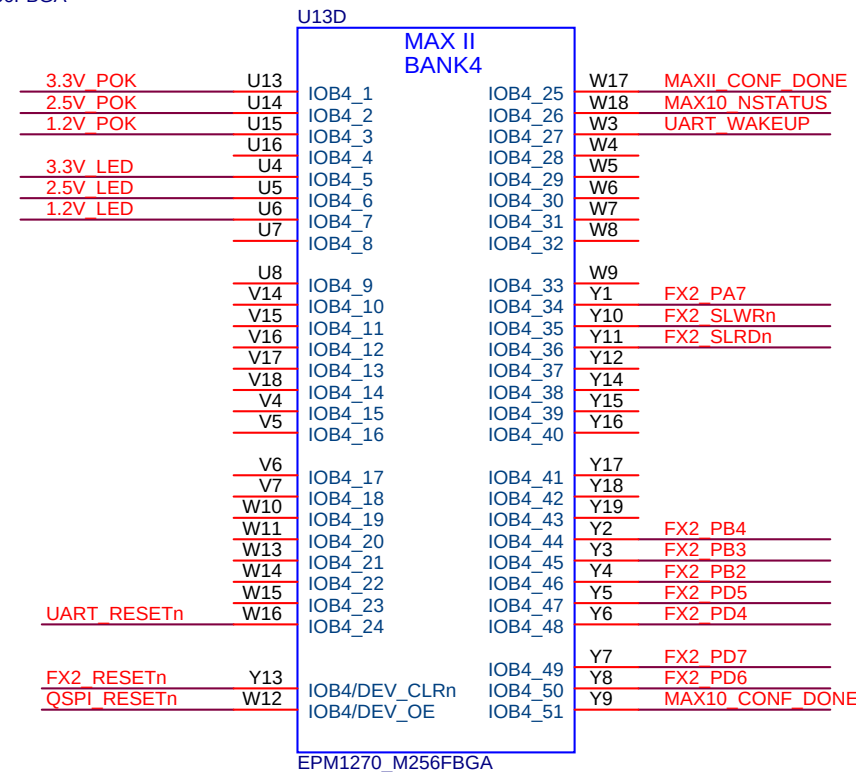
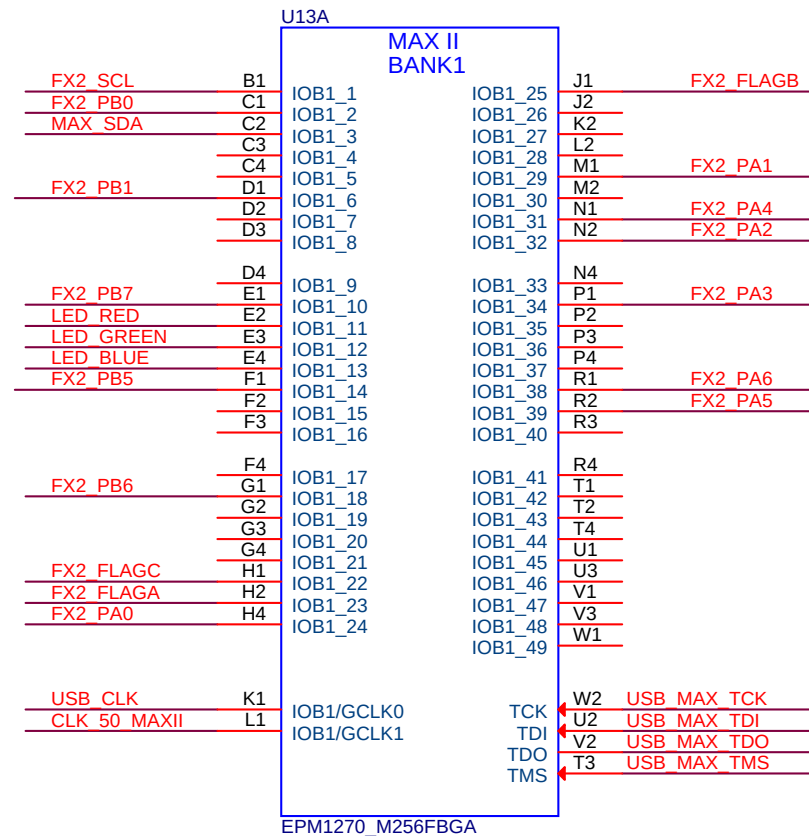
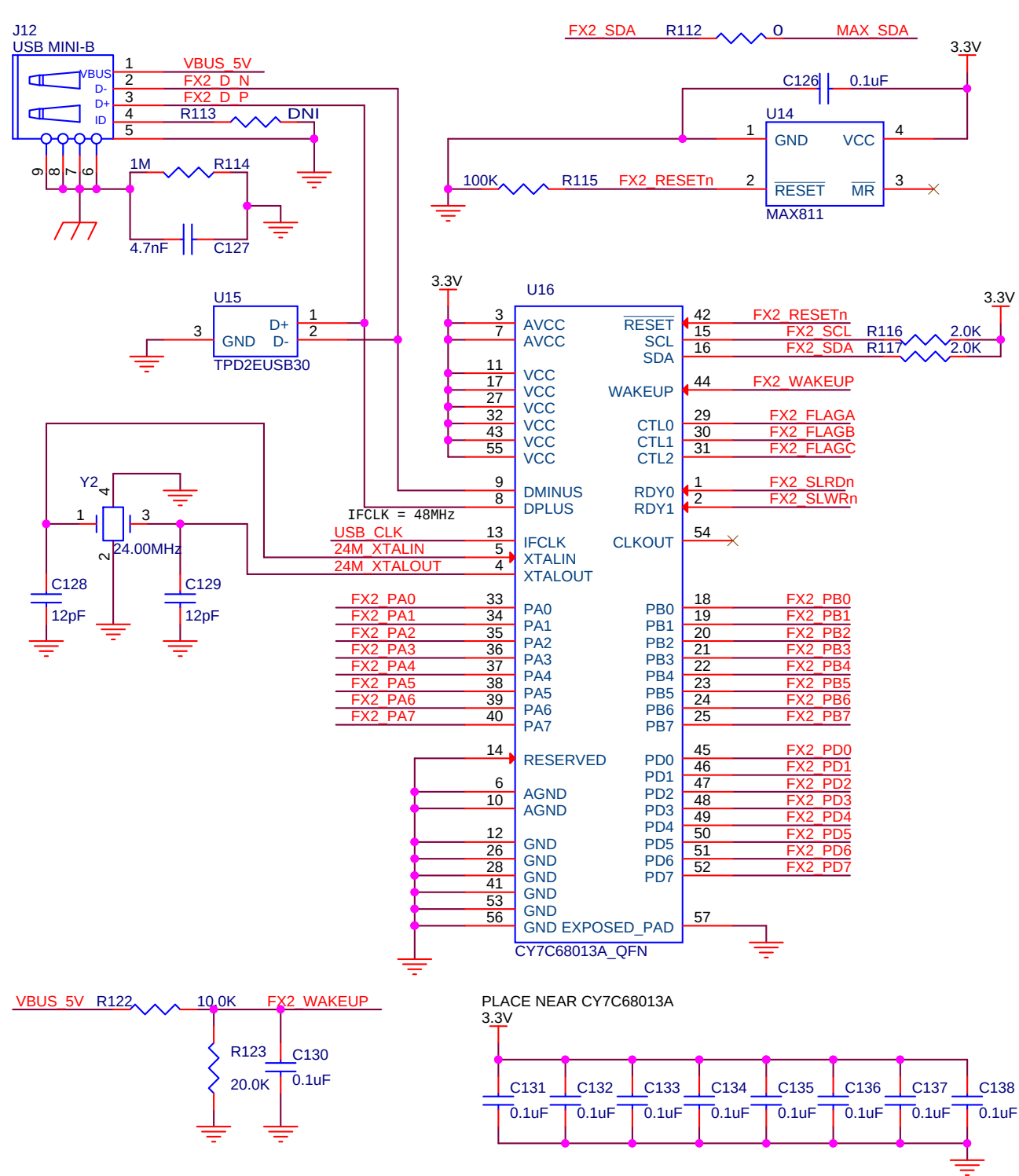


2.5V default 3.3V optional



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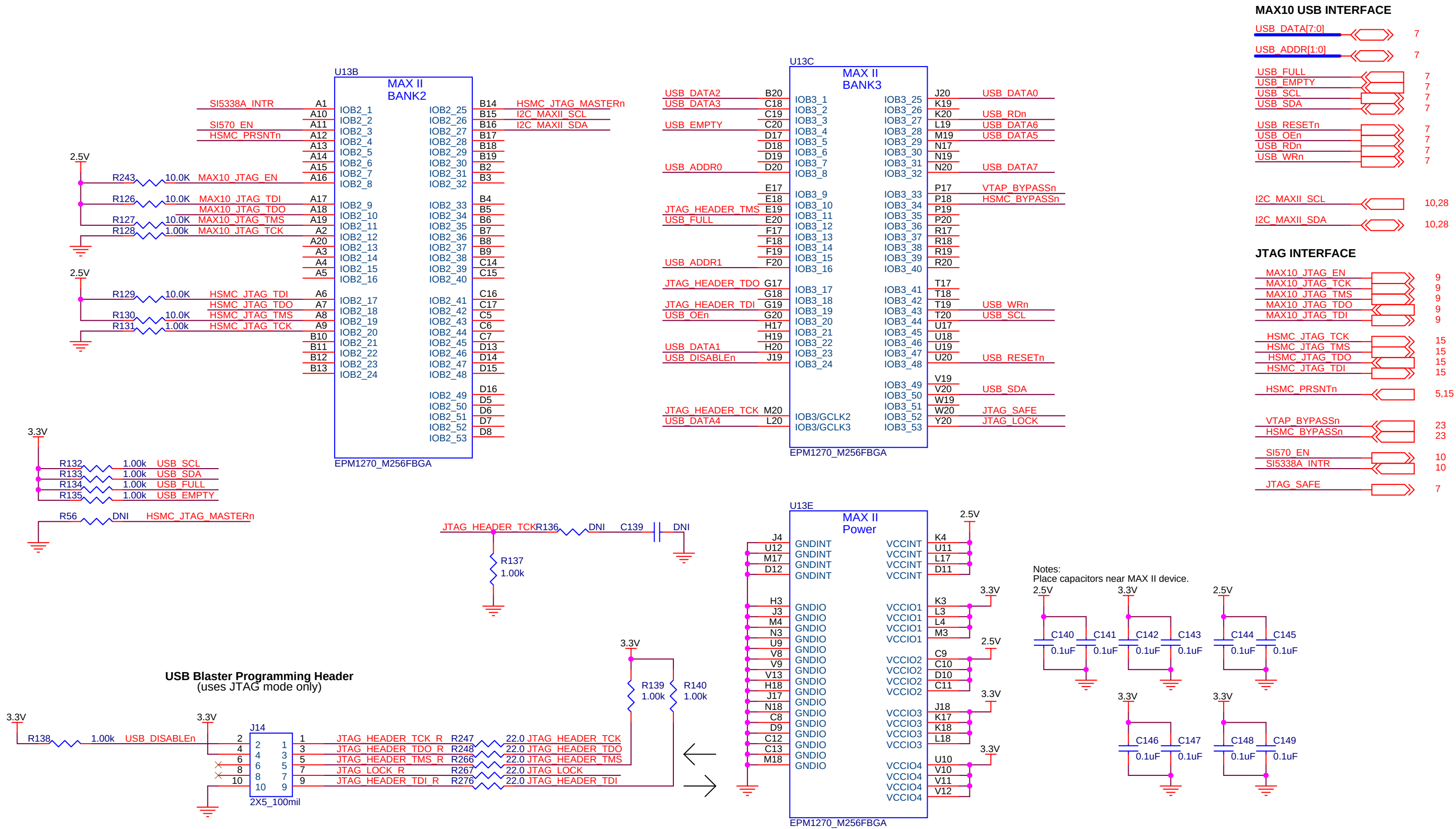
On-Board USB Blaster II-1



altera™

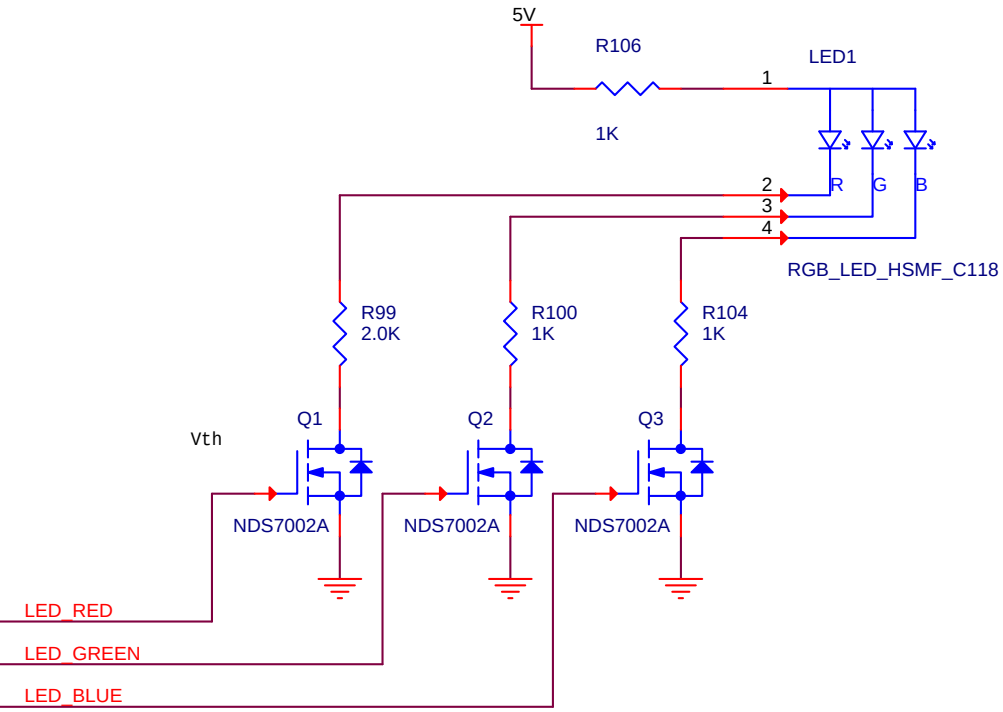
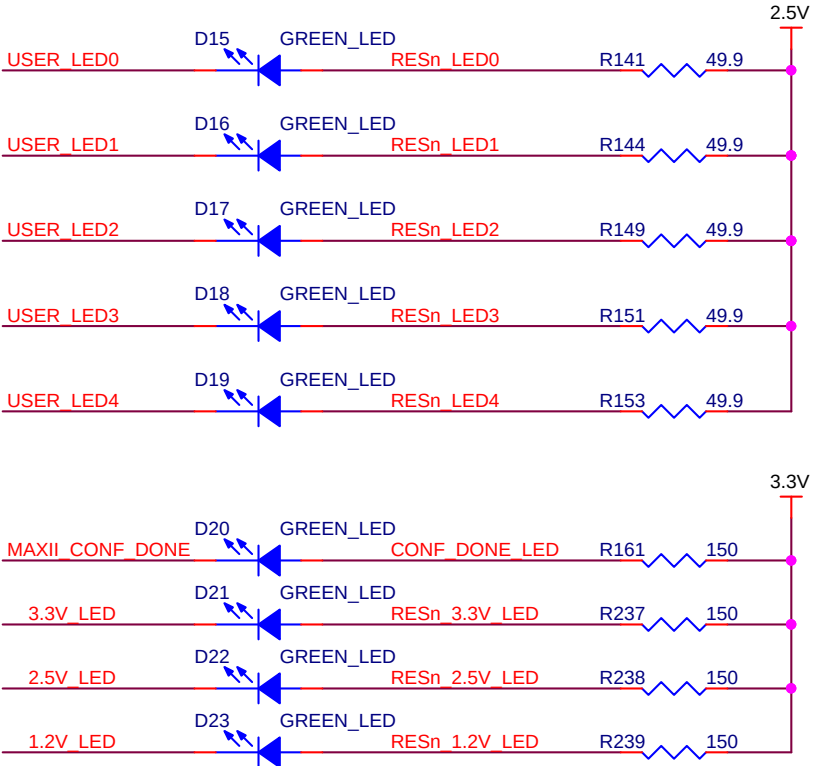
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On-Board USB Blaster II-2

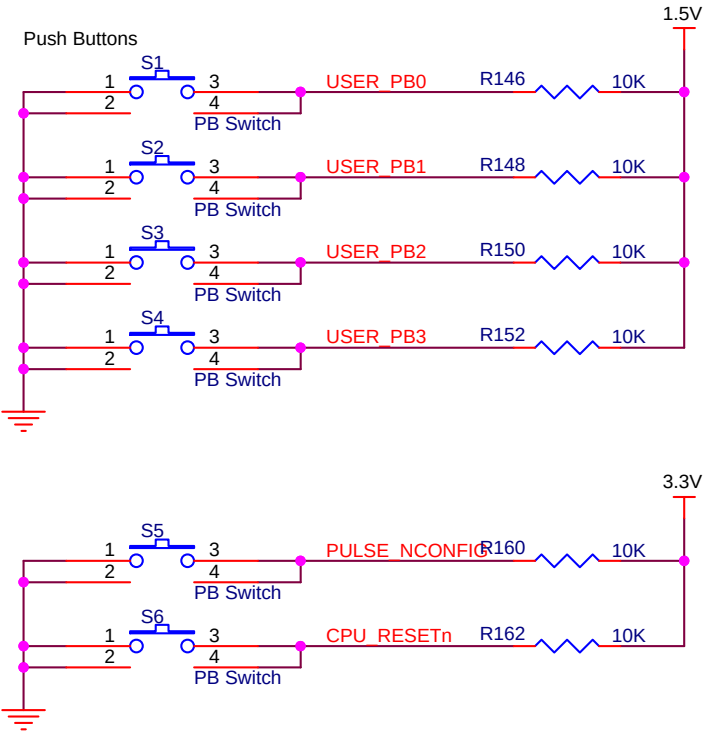
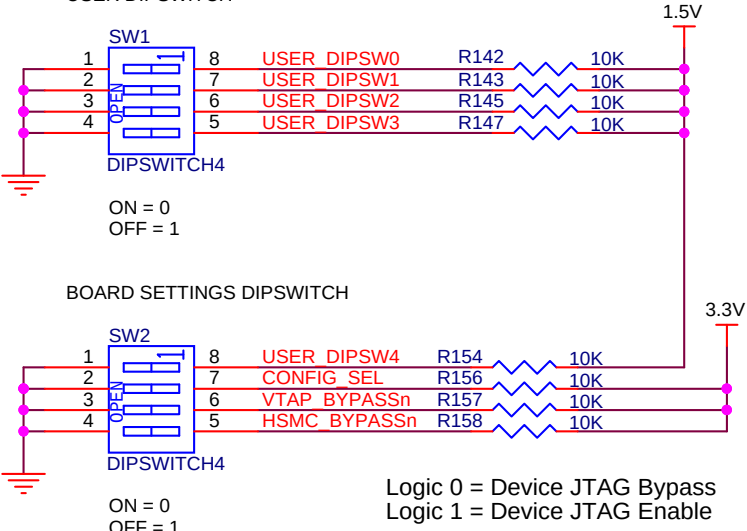


LED, User IO, Connector

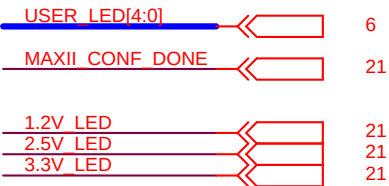
LED



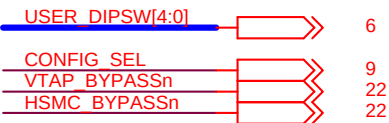
USER DIPSWITCH



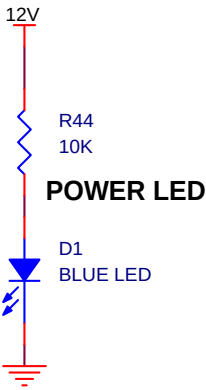
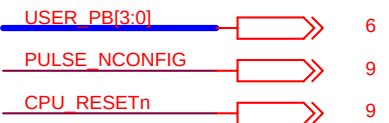
LED



Switches

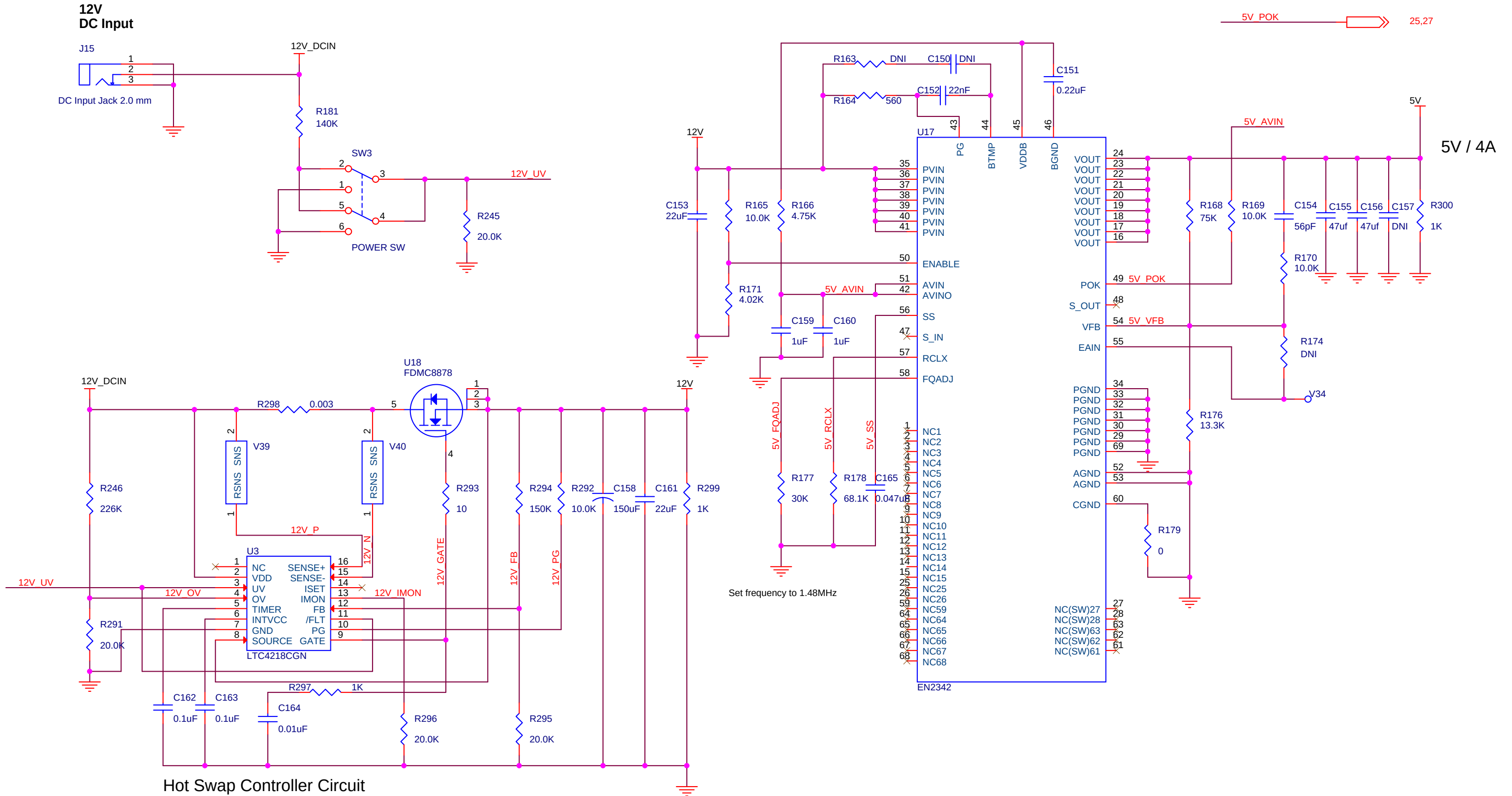


Buttons

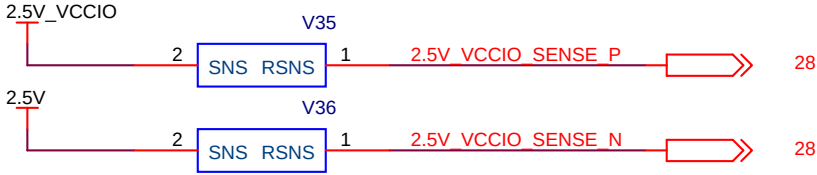
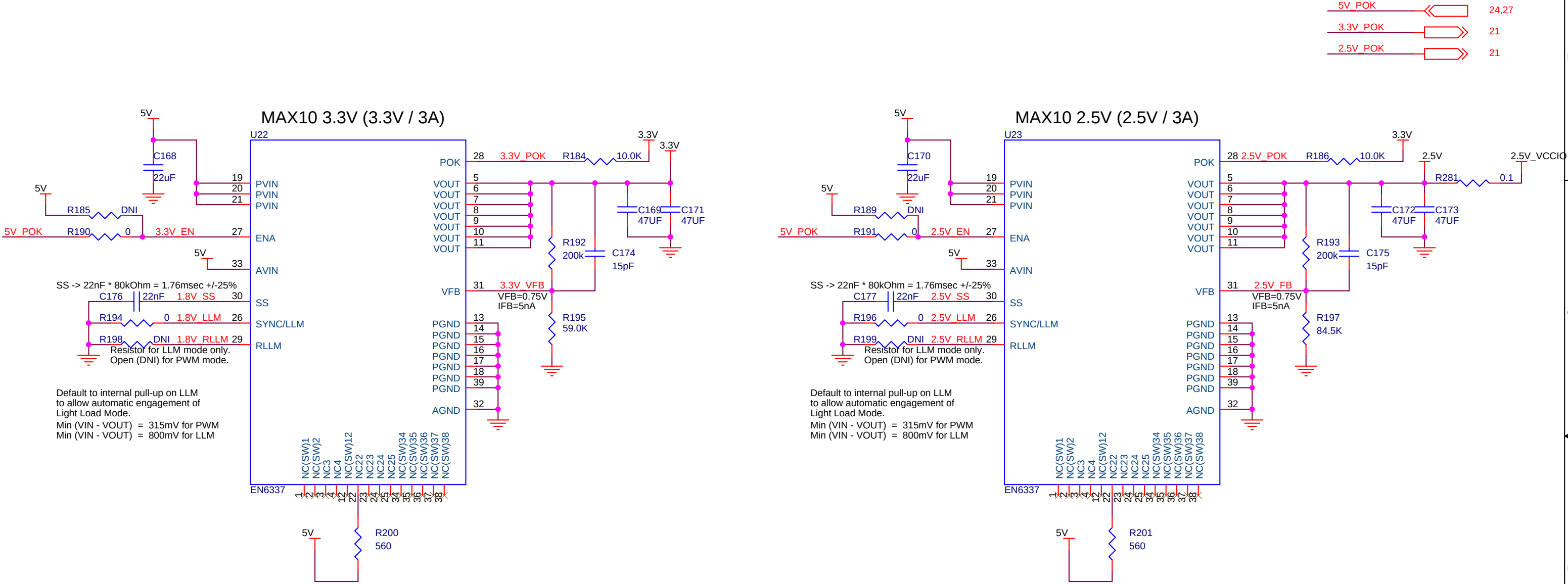


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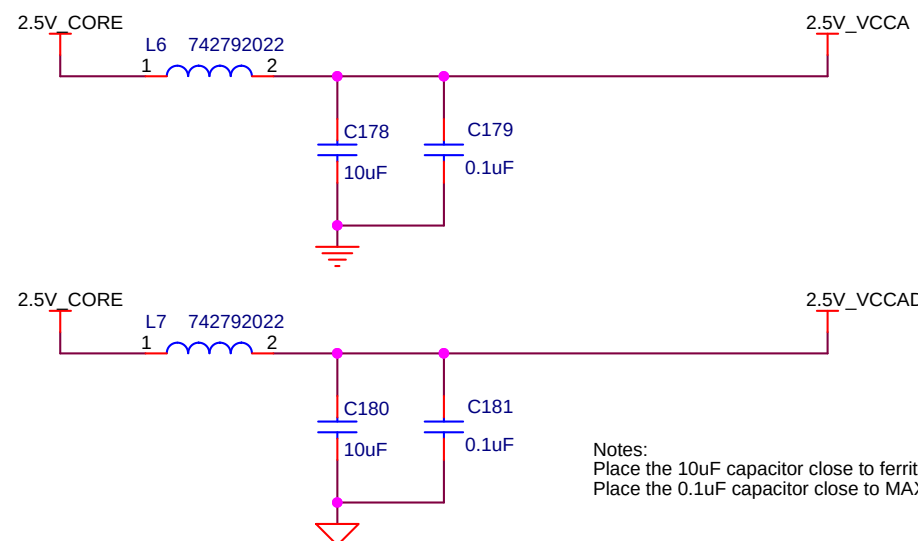
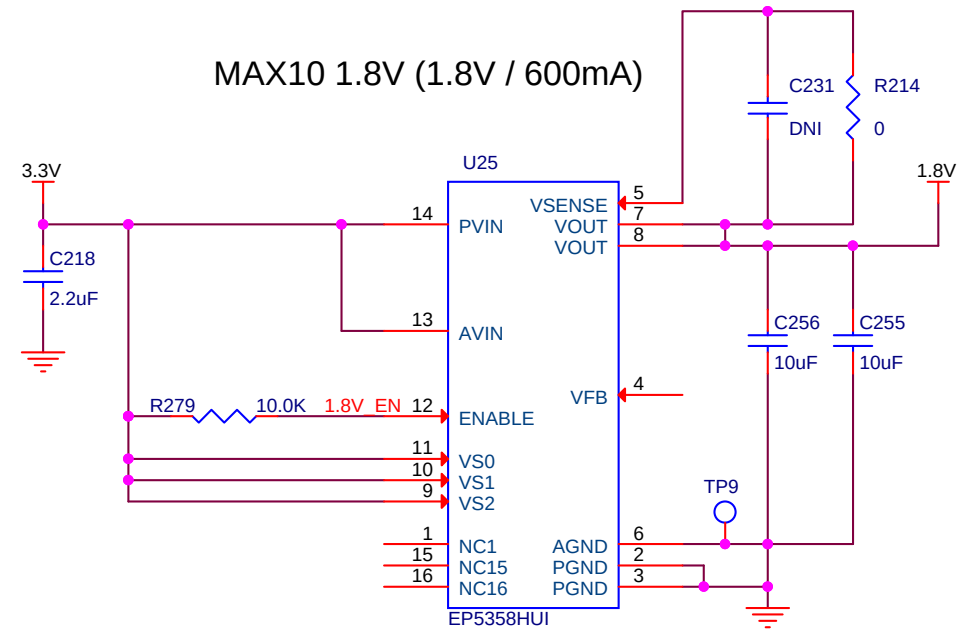
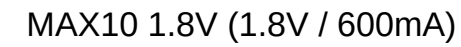
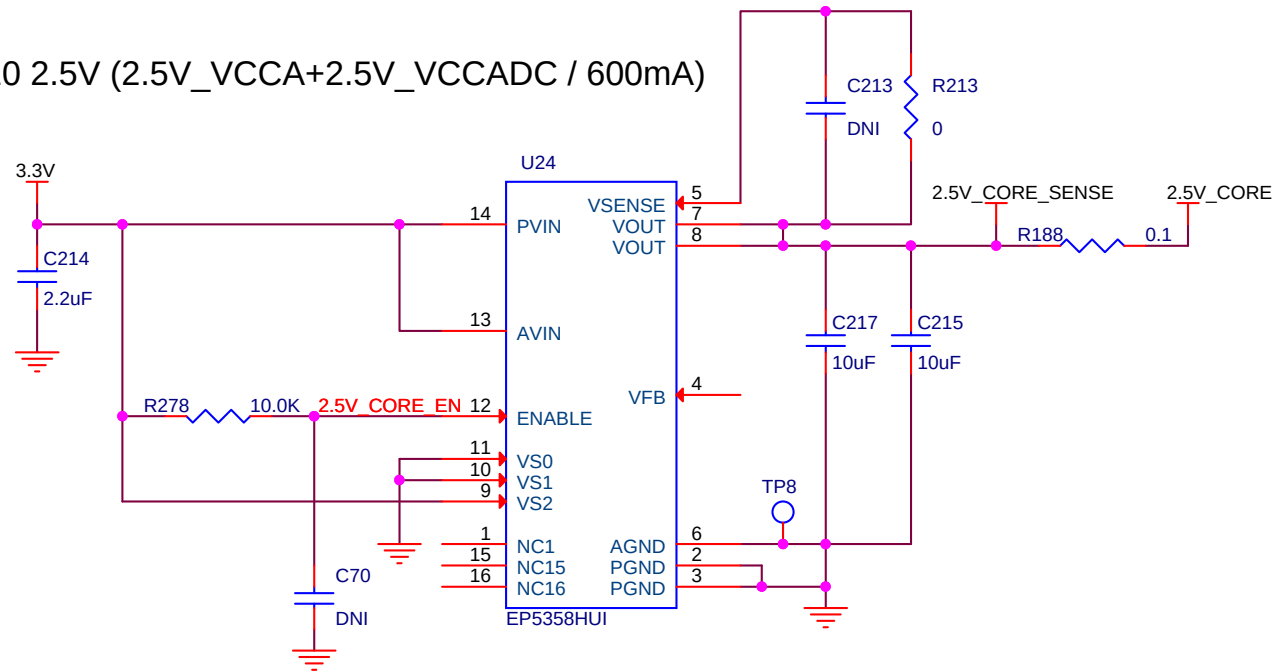
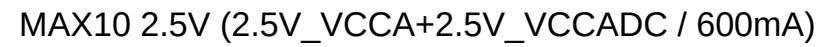
Power - 12V_DCIN / 5V



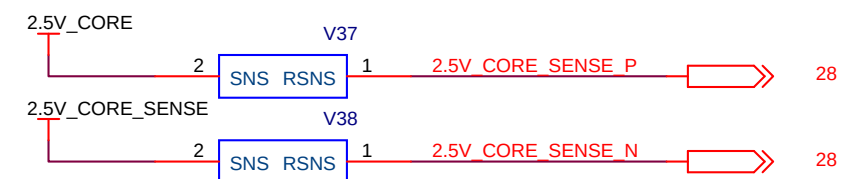
Power - 3.3V / 2.5V



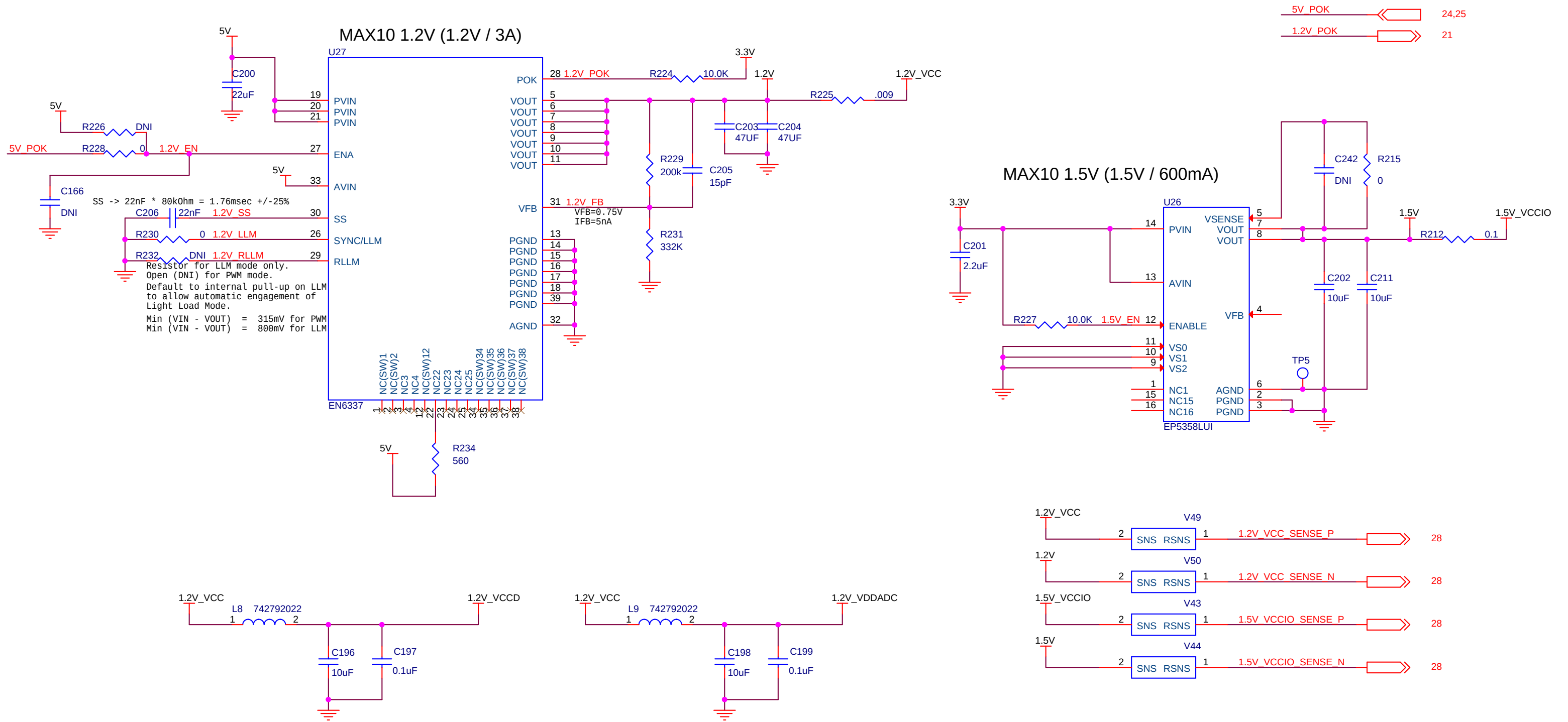
Power - 2.5V_VCC / 1.2V



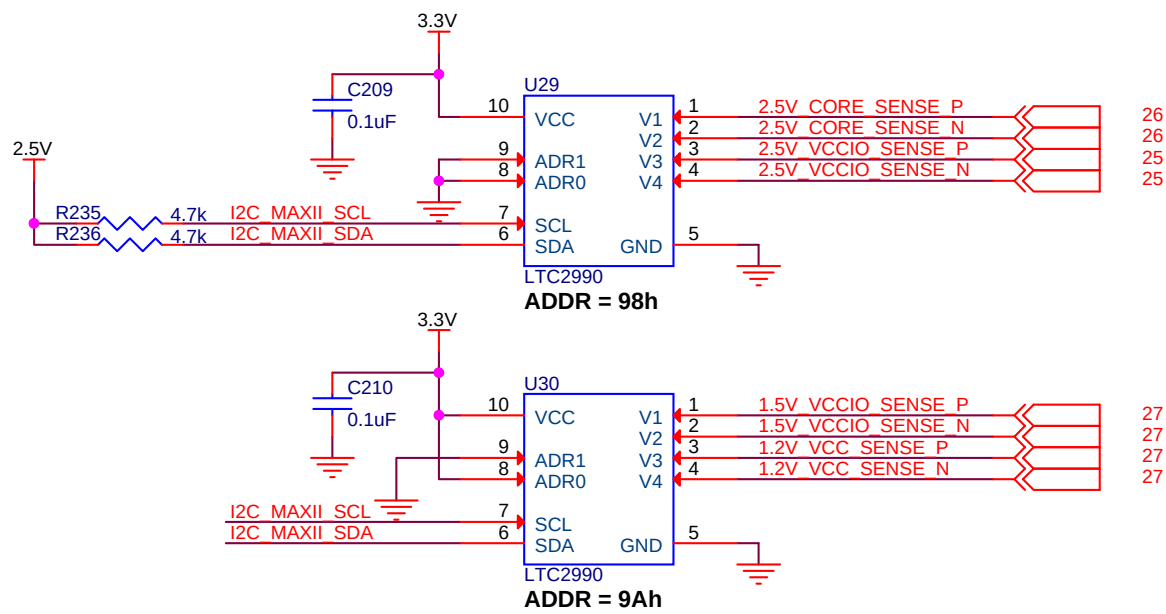
Notes:
Place the 10uF capacitor close to ferrite bead.
Place the 0.1uF capacitor close to MAX 10 pin.



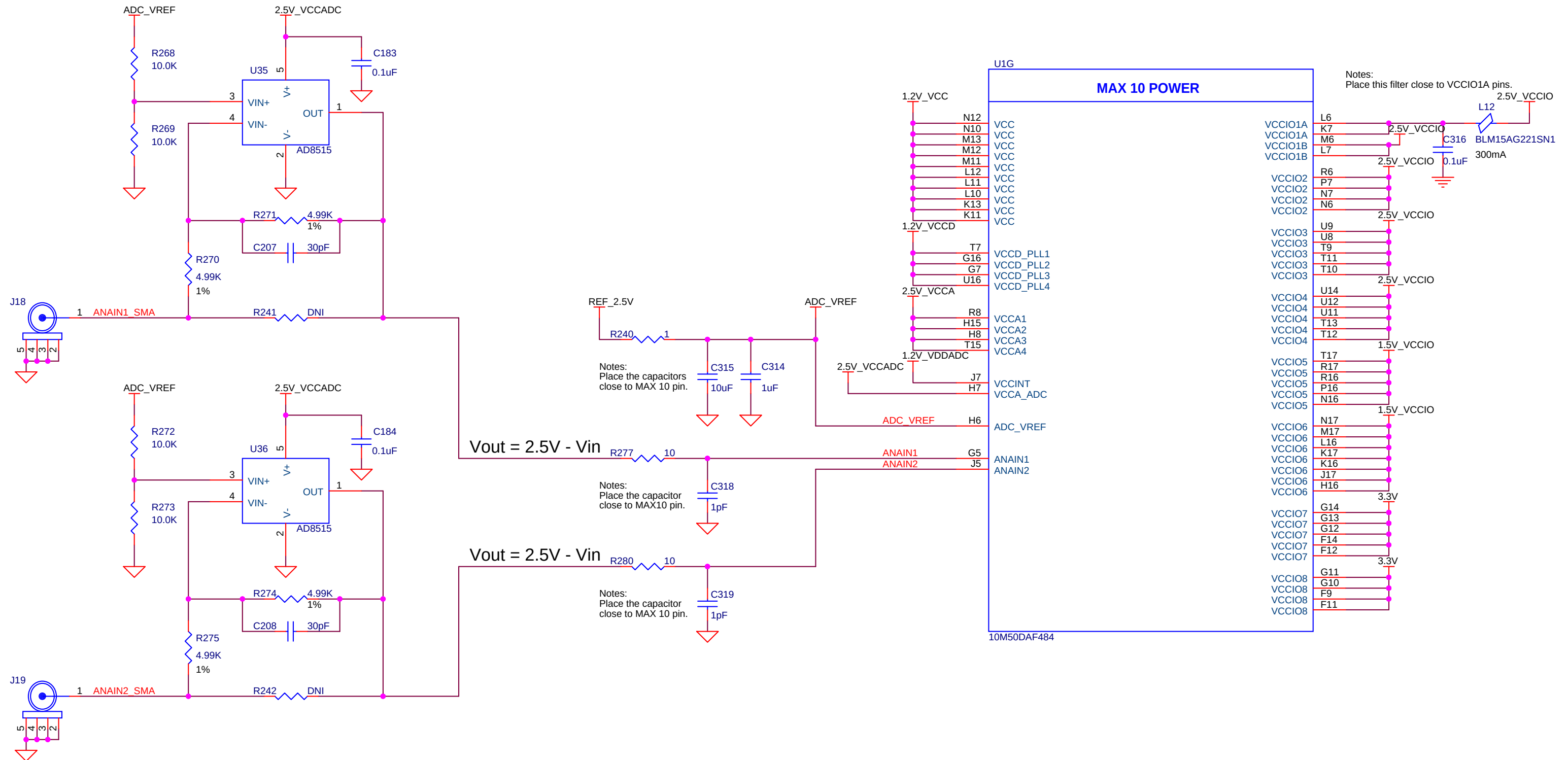
Power - 1.5V / 1.2V



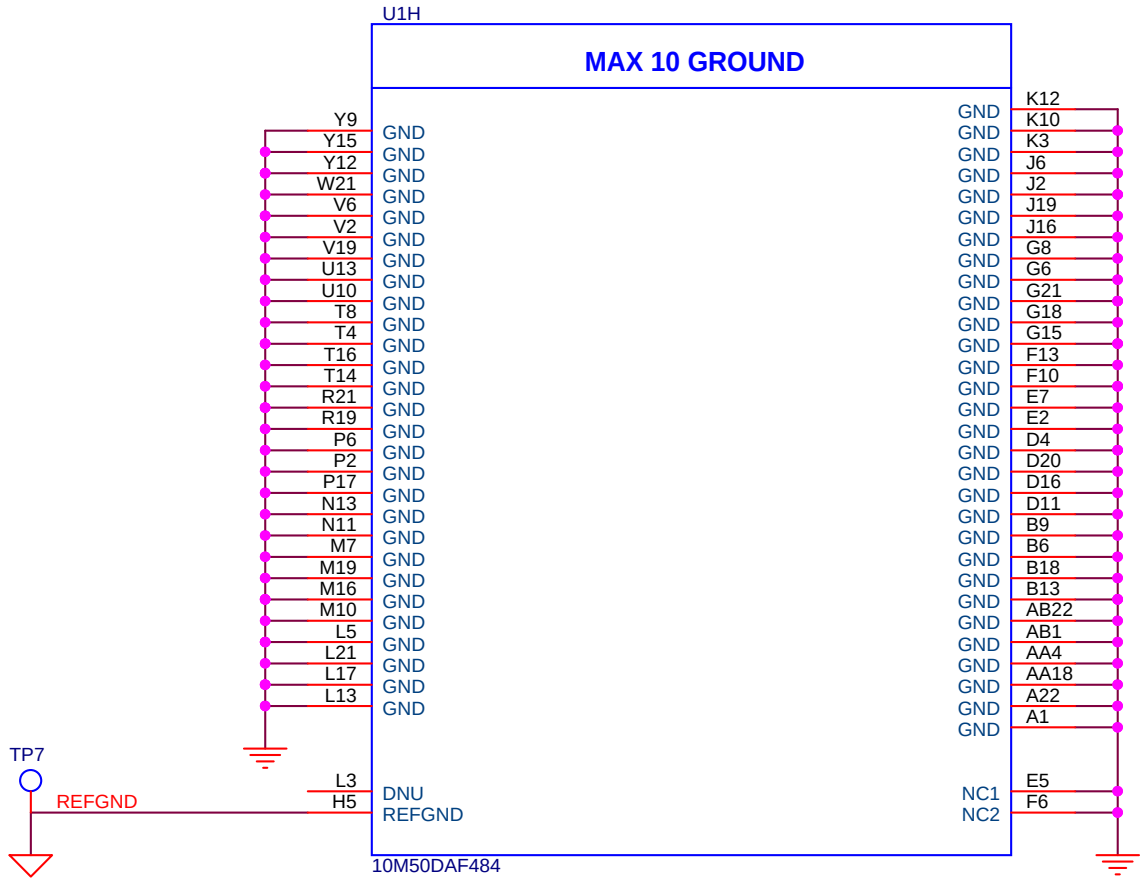
Power - Power Monitor



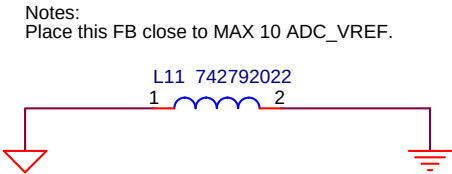
MAX10 Power



MAX10 Ground

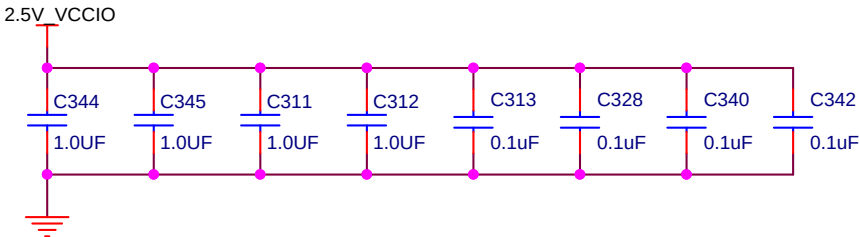
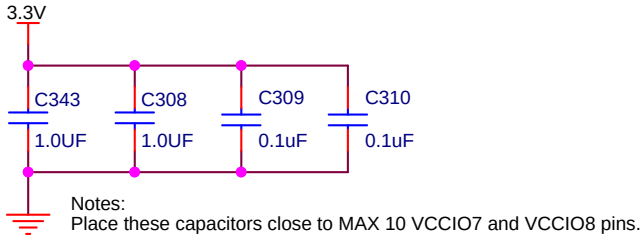
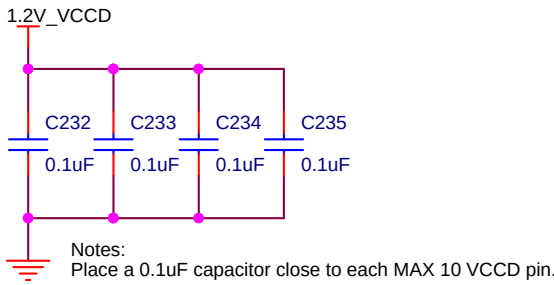
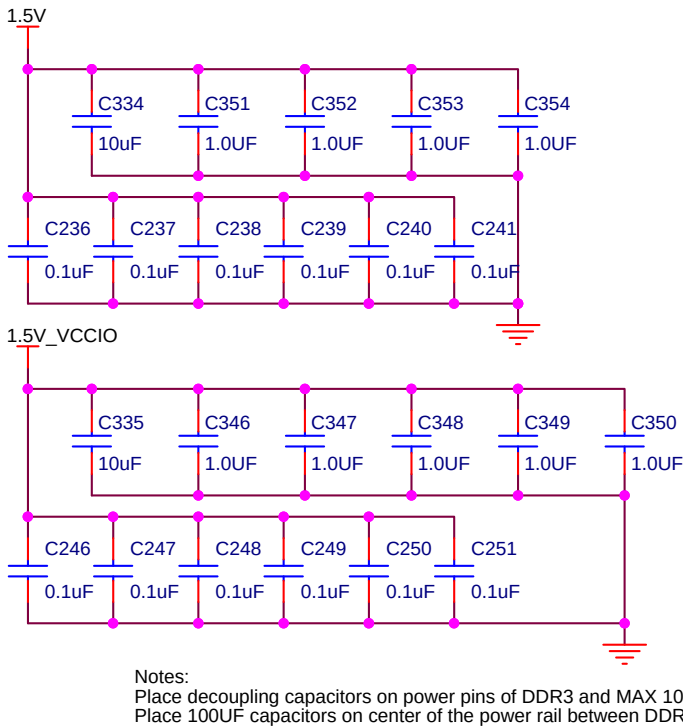
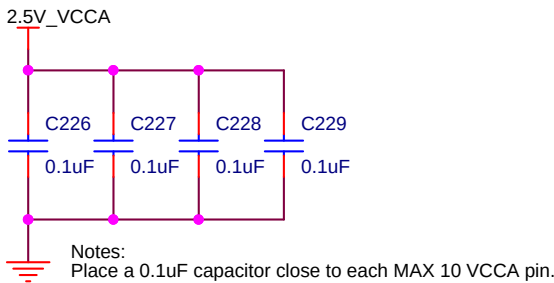
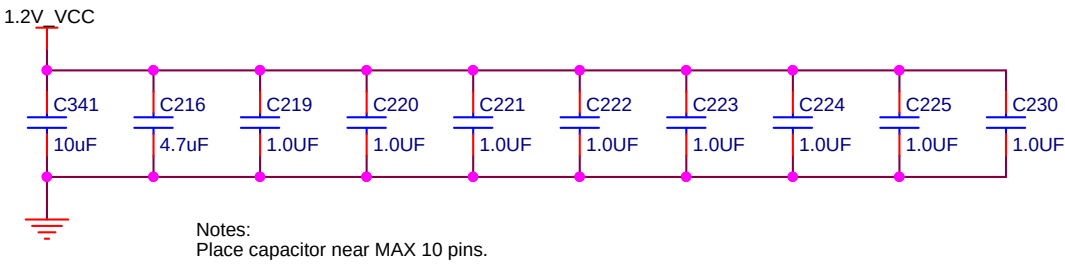


Notes:
1. Use REFVREF as ground reference.
2. Route analog input signal adjacent to AVSSREF as possible.



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MAX10 Decoupling



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